

CHAPTER 6: INNER PRODUCT SPACES

6.1 Inner Products

1.
 - (a) $\langle \mathbf{u}, \mathbf{v} \rangle = 2(1)(3) + 3(1)(2) = 12$
 - (b) $\langle k\mathbf{v}, \mathbf{w} \rangle = 2((3)(3))(0) + 3((3)(2))(-1) = -18$
 - (c) $\langle \mathbf{u} + \mathbf{v}, \mathbf{w} \rangle = 2(1+3)(0) + 3(1+2)(-1) = -9$
 - (d) $\|\mathbf{v}\| = \langle \mathbf{v}, \mathbf{v} \rangle^{1/2} = [2(3)(3) + 3(2)(2)]^{1/2} = \sqrt{30}$
 - (e) $d(\mathbf{u}, \mathbf{v}) = \|\mathbf{u} - \mathbf{v}\| = \langle (-2, -1), (-2, -1) \rangle^{1/2} = [2(-2)(-2) + 3(-1)(-1)]^{1/2} = \sqrt{11}$
 - (f) $\|\mathbf{u} - k\mathbf{v}\| = \langle (-8, -5), (-8, -5) \rangle^{1/2} = [2(-8)(-8) + 3(-5)(-5)]^{1/2} = \sqrt{203}$
2.
 - (a) $\langle \mathbf{u}, \mathbf{v} \rangle = \frac{1}{2}(1)(3) + 5(1)(2) = \frac{23}{2}$
 - (b) $\langle k\mathbf{v}, \mathbf{w} \rangle = \frac{1}{2}((3)(3))(0) + 5((3)(2))(-1) = -30$
 - (c) $\langle \mathbf{u} + \mathbf{v}, \mathbf{w} \rangle = \frac{1}{2}(1+3)(0) + 5(1+2)(-1) = -15$
 - (d) $\|\mathbf{v}\| = \langle \mathbf{v}, \mathbf{v} \rangle^{1/2} = \left[\frac{1}{2}(3)(3) + 5(2)(2) \right]^{1/2} = \sqrt{\frac{49}{2}} = \frac{7}{\sqrt{2}}$
 - (e) $d(\mathbf{u}, \mathbf{v}) = \|\mathbf{u} - \mathbf{v}\| = \langle (-2, -1), (-2, -1) \rangle^{1/2} = \left[\frac{1}{2}(-2)(-2) + 5(-1)(-1) \right]^{1/2} = \sqrt{7}$
 - (f) $\|\mathbf{u} - k\mathbf{v}\| = \langle (-8, -5), (-8, -5) \rangle^{1/2} = \left[\frac{1}{2}(-8)(-8) + 5(-5)(-5) \right]^{1/2} = \sqrt{157}$
3.
 - (a) $\langle \mathbf{u}, \mathbf{v} \rangle = \left(\begin{bmatrix} 2 & 1 \\ 1 & 1 \end{bmatrix} \begin{bmatrix} 1 \\ 1 \end{bmatrix} \right) \cdot \left(\begin{bmatrix} 2 & 1 \\ 1 & 1 \end{bmatrix} \begin{bmatrix} 3 \\ 2 \end{bmatrix} \right) = \begin{bmatrix} 3 \\ 2 \end{bmatrix} \cdot \begin{bmatrix} 8 \\ 5 \end{bmatrix} = 34$
 - (b) $\langle k\mathbf{v}, \mathbf{w} \rangle = \left(\begin{bmatrix} 2 & 1 \\ 1 & 1 \end{bmatrix} \begin{bmatrix} 9 \\ 6 \end{bmatrix} \right) \cdot \left(\begin{bmatrix} 2 & 1 \\ 1 & 1 \end{bmatrix} \begin{bmatrix} 0 \\ -1 \end{bmatrix} \right) = \begin{bmatrix} 24 \\ 15 \end{bmatrix} \cdot \begin{bmatrix} -1 \\ -1 \end{bmatrix} = -39$
 - (c) $\langle \mathbf{u} + \mathbf{v}, \mathbf{w} \rangle = \left(\begin{bmatrix} 2 & 1 \\ 1 & 1 \end{bmatrix} \begin{bmatrix} 4 \\ 3 \end{bmatrix} \right) \cdot \left(\begin{bmatrix} 2 & 1 \\ 1 & 1 \end{bmatrix} \begin{bmatrix} 0 \\ -1 \end{bmatrix} \right) = \begin{bmatrix} 11 \\ 7 \end{bmatrix} \cdot \begin{bmatrix} -1 \\ -1 \end{bmatrix} = -18$
 - (d) $\|\mathbf{v}\| = \langle \mathbf{v}, \mathbf{v} \rangle^{1/2} = \left[\left(\begin{bmatrix} 2 & 1 \\ 1 & 1 \end{bmatrix} \begin{bmatrix} 3 \\ 2 \end{bmatrix} \right) \cdot \left(\begin{bmatrix} 2 & 1 \\ 1 & 1 \end{bmatrix} \begin{bmatrix} 3 \\ 2 \end{bmatrix} \right) \right]^{1/2} = \left(\begin{bmatrix} 8 \\ 5 \end{bmatrix} \cdot \begin{bmatrix} 8 \\ 5 \end{bmatrix} \right)^{1/2} = \sqrt{89}$
 - (e) $d(\mathbf{u}, \mathbf{v}) = \|\mathbf{u} - \mathbf{v}\| = \left[\left(\begin{bmatrix} 2 & 1 \\ 1 & 1 \end{bmatrix} \begin{bmatrix} -2 \\ -1 \end{bmatrix} \right) \cdot \left(\begin{bmatrix} 2 & 1 \\ 1 & 1 \end{bmatrix} \begin{bmatrix} -2 \\ -1 \end{bmatrix} \right) \right]^{1/2} = \left(\begin{bmatrix} -5 \\ -3 \end{bmatrix} \cdot \begin{bmatrix} -5 \\ -3 \end{bmatrix} \right)^{1/2} = \sqrt{34}$

$$(f) \quad \|\mathbf{u} - k\mathbf{v}\| = \left[\left(\begin{bmatrix} 2 & 1 \\ 1 & 1 \end{bmatrix} \begin{bmatrix} -8 \\ -5 \end{bmatrix} \right) \cdot \left(\begin{bmatrix} 2 & 1 \\ 1 & 1 \end{bmatrix} \begin{bmatrix} -8 \\ -5 \end{bmatrix} \right) \right]^{1/2} = \left(\begin{bmatrix} -21 \\ -13 \end{bmatrix} \cdot \begin{bmatrix} -21 \\ -13 \end{bmatrix} \right)^{1/2} = \sqrt{610}$$

$$4. \quad (a) \quad \langle \mathbf{u}, \mathbf{v} \rangle = \left(\begin{bmatrix} 1 & 0 \\ 2 & -1 \end{bmatrix} \begin{bmatrix} 1 \\ 1 \end{bmatrix} \right) \cdot \left(\begin{bmatrix} 1 & 0 \\ 2 & -1 \end{bmatrix} \begin{bmatrix} 3 \\ 2 \end{bmatrix} \right) = \begin{bmatrix} 1 \\ 1 \end{bmatrix} \cdot \begin{bmatrix} 3 \\ 4 \end{bmatrix} = 7$$

$$(b) \quad \langle k\mathbf{v}, \mathbf{w} \rangle = \left(\begin{bmatrix} 1 & 0 \\ 2 & -1 \end{bmatrix} \begin{bmatrix} 9 \\ 6 \end{bmatrix} \right) \cdot \left(\begin{bmatrix} 1 & 0 \\ 2 & -1 \end{bmatrix} \begin{bmatrix} 0 \\ -1 \end{bmatrix} \right) = \begin{bmatrix} 9 \\ 12 \end{bmatrix} \cdot \begin{bmatrix} 0 \\ 1 \end{bmatrix} = 12$$

$$(c) \quad \langle \mathbf{u} + \mathbf{v}, \mathbf{w} \rangle = \left(\begin{bmatrix} 1 & 0 \\ 2 & -1 \end{bmatrix} \begin{bmatrix} 4 \\ 3 \end{bmatrix} \right) \cdot \left(\begin{bmatrix} 1 & 0 \\ 2 & -1 \end{bmatrix} \begin{bmatrix} 0 \\ -1 \end{bmatrix} \right) = \begin{bmatrix} 4 \\ 5 \end{bmatrix} \cdot \begin{bmatrix} 0 \\ 1 \end{bmatrix} = 5$$

$$(d) \quad \|\mathbf{v}\| = \langle \mathbf{v}, \mathbf{v} \rangle^{1/2} = \left[\left(\begin{bmatrix} 1 & 0 \\ 2 & -1 \end{bmatrix} \begin{bmatrix} 3 \\ 2 \end{bmatrix} \right) \cdot \left(\begin{bmatrix} 1 & 0 \\ 2 & -1 \end{bmatrix} \begin{bmatrix} 3 \\ 2 \end{bmatrix} \right) \right]^{1/2} = \left(\begin{bmatrix} 3 \\ 4 \end{bmatrix} \cdot \begin{bmatrix} 3 \\ 4 \end{bmatrix} \right)^{1/2} = \sqrt{25} = 5$$

$$(e) \quad d(\mathbf{u}, \mathbf{v}) = \|\mathbf{u} - \mathbf{v}\| = \left[\left(\begin{bmatrix} 1 & 0 \\ 2 & -1 \end{bmatrix} \begin{bmatrix} -2 \\ -1 \end{bmatrix} \right) \cdot \left(\begin{bmatrix} 1 & 0 \\ 2 & -1 \end{bmatrix} \begin{bmatrix} -2 \\ -1 \end{bmatrix} \right) \right]^{1/2} = \left(\begin{bmatrix} -2 \\ -3 \end{bmatrix} \cdot \begin{bmatrix} -2 \\ -3 \end{bmatrix} \right)^{1/2} = \sqrt{13}$$

$$(f) \quad \|\mathbf{u} - k\mathbf{v}\| = \left[\left(\begin{bmatrix} 1 & 0 \\ 2 & -1 \end{bmatrix} \begin{bmatrix} -8 \\ -5 \end{bmatrix} \right) \cdot \left(\begin{bmatrix} 1 & 0 \\ 2 & -1 \end{bmatrix} \begin{bmatrix} -8 \\ -5 \end{bmatrix} \right) \right]^{1/2} = \left(\begin{bmatrix} -8 \\ -11 \end{bmatrix} \cdot \begin{bmatrix} -8 \\ -11 \end{bmatrix} \right)^{1/2} = \sqrt{185}$$

$$5. \quad \begin{bmatrix} \sqrt{2} & 0 \\ 0 & \sqrt{3} \end{bmatrix}$$

$$6. \quad \begin{bmatrix} \frac{1}{\sqrt{2}} & 0 \\ 0 & \sqrt{5} \end{bmatrix}$$

$$7. \quad \langle \mathbf{u}, \mathbf{v} \rangle = \left(\begin{bmatrix} 4 & 1 \\ 2 & -3 \end{bmatrix} \begin{bmatrix} 0 \\ -3 \end{bmatrix} \right) \cdot \left(\begin{bmatrix} 4 & 1 \\ 2 & -3 \end{bmatrix} \begin{bmatrix} 6 \\ 2 \end{bmatrix} \right) = \begin{bmatrix} -3 \\ 9 \end{bmatrix} \cdot \begin{bmatrix} 26 \\ 6 \end{bmatrix} = -24$$

$$8. \quad \langle \mathbf{u}, \mathbf{v} \rangle = \left(\begin{bmatrix} 2 & 1 \\ -1 & 3 \end{bmatrix} \begin{bmatrix} 0 \\ -3 \end{bmatrix} \right) \cdot \left(\begin{bmatrix} 2 & 1 \\ -1 & 3 \end{bmatrix} \begin{bmatrix} 6 \\ 2 \end{bmatrix} \right) = \begin{bmatrix} -3 \\ -9 \end{bmatrix} \cdot \begin{bmatrix} 14 \\ 0 \end{bmatrix} = -42$$

$$9. \quad \text{If } \mathbf{u} = U \text{ and } \mathbf{v} = V \text{ then } \langle \mathbf{u}, \mathbf{v} \rangle = \text{tr}(U^T V) = \text{tr} \left(\begin{bmatrix} 1 & 13 \\ 10 & 2 \end{bmatrix} \right) = 3.$$

$$10. \quad \text{If } \mathbf{u} = U \text{ and } \mathbf{v} = V \text{ then } \langle \mathbf{u}, \mathbf{v} \rangle = \text{tr}(U^T V) = \text{tr} \left(\begin{bmatrix} 4 & -18 \\ 8 & 52 \end{bmatrix} \right) = 56.$$

$$11. \quad \langle \mathbf{p}, \mathbf{q} \rangle = (-2)(4) + (1)(0) + (3)(-7) = -29$$

$$12. \quad \langle \mathbf{p}, \mathbf{q} \rangle = (-5)(3) + (2)(2) + (1)(-4) = -15$$

$$13. \begin{bmatrix} \sqrt{3} & 0 \\ 0 & \sqrt{5} \end{bmatrix}$$

$$14. \begin{bmatrix} 2 & 0 \\ 0 & \sqrt{6} \end{bmatrix}$$

$$15. \begin{aligned} \langle \mathbf{p}, \mathbf{q} \rangle &= p(-2)q(-2) + p(-1)q(-1) + p(0)q(0) + p(1)q(1) \\ &= (-10)(5) + (-2)(2) + (0)(1) + (2)(2) = -50 \end{aligned}$$

$$16. \begin{aligned} \langle \mathbf{p}, \mathbf{q} \rangle &= p(-1)q(-1) + p(0)q(0) + p(1)q(1) + p(2)q(2) \\ &= (-2)(2) + (0)(1) + (2)(2) + (10)(5) = 50 \end{aligned}$$

$$17. \begin{aligned} \|\mathbf{u}\| &= \langle \mathbf{u}, \mathbf{u} \rangle^{1/2} = [2(-3)(-3) + 3(2)(2)]^{1/2} = \sqrt{30} \\ d(\mathbf{u}, \mathbf{v}) &= \|\mathbf{u} - \mathbf{v}\| = \langle (-4, -5), (-4, -5) \rangle^{1/2} = [2(-4)(-4) + 3(-5)(-5)]^{1/2} = \sqrt{107} \end{aligned}$$

$$18. \begin{aligned} \|\mathbf{u}\| &= \langle \mathbf{u}, \mathbf{u} \rangle^{1/2} = [2(-1)(-1) + 3(2)(2)]^{1/2} = \sqrt{14} \\ d(\mathbf{u}, \mathbf{v}) &= \|\mathbf{u} - \mathbf{v}\| = \langle (-3, -3), (-3, -3) \rangle^{1/2} = [2(-3)(-3) + 3(-3)(-3)]^{1/2} = 3\sqrt{5} \end{aligned}$$

$$19. \|\mathbf{p}\| = \langle \mathbf{p}, \mathbf{p} \rangle^{1/2} = \sqrt{(-2)^2 + 1^2 + 3^2} = \sqrt{14}; \quad d(\mathbf{p}, \mathbf{q}) = \|\mathbf{p} - \mathbf{q}\| = \sqrt{(-6)^2 + 1^2 + 10^2} = \sqrt{137}$$

$$20. \|\mathbf{p}\| = \langle \mathbf{p}, \mathbf{p} \rangle^{1/2} = \sqrt{(-5)^2 + 2^2 + 1^2} = \sqrt{30}; \quad d(\mathbf{p}, \mathbf{q}) = \|\mathbf{p} - \mathbf{q}\| = \sqrt{(-8)^2 + 0^2 + 5^2} = \sqrt{89}$$

$$21. \text{ If } \mathbf{u} = U \text{ and } \mathbf{v} = V \text{ then } \|U\| = \langle \mathbf{u}, \mathbf{u} \rangle^{1/2} = \sqrt{\text{tr}(U^T U)} = \sqrt{\text{tr}\left(\begin{bmatrix} 25 & 26 \\ 26 & 68 \end{bmatrix}\right)} = \sqrt{93} \text{ and}$$

$$d(U, V) = \|U - V\| = \langle \mathbf{u} - \mathbf{v}, \mathbf{u} - \mathbf{v} \rangle^{1/2} = \sqrt{\text{tr}((U - V)^T (U - V))} = \sqrt{\text{tr}\left(\begin{bmatrix} 25 & 1 \\ 1 & 74 \end{bmatrix}\right)} = \sqrt{99} = 3\sqrt{11}.$$

$$22. \text{ If } \mathbf{u} = U \text{ and } \mathbf{v} = V \text{ then } \|U\| = \langle \mathbf{u}, \mathbf{u} \rangle^{1/2} = \sqrt{\text{tr}(U^T U)} = \sqrt{\text{tr}\left(\begin{bmatrix} 10 & -13 \\ -13 & 29 \end{bmatrix}\right)} = \sqrt{39} \text{ and}$$

$$d(U, V) = \|U - V\| = \langle \mathbf{u} - \mathbf{v}, \mathbf{u} - \mathbf{v} \rangle^{1/2} = \sqrt{\text{tr}((U - V)^T (U - V))} = \sqrt{\text{tr}\left(\begin{bmatrix} 18 & 21 \\ 21 & 25 \end{bmatrix}\right)} = \sqrt{43}.$$

$$23. \begin{aligned} \|\mathbf{p}\| &= \langle \mathbf{p}, \mathbf{p} \rangle^{1/2} = \sqrt{[p(-2)]^2 + [p(-1)]^2 + [p(0)]^2 + [p(1)]^2} = \sqrt{(-10)^2 + (-2)^2 + 0^2 + 2^2} = 6\sqrt{3} \\ d(\mathbf{p}, \mathbf{q}) &= \|\mathbf{p} - \mathbf{q}\| = \sqrt{[p(-2) - q(-2)]^2 + [p(-1) - q(-1)]^2 + [p(0) - q(0)]^2 + [p(1) - q(1)]^2} \\ &= \sqrt{(-15)^2 + (-4)^2 + (-1)^2 + 0^2} = 11\sqrt{2} \end{aligned}$$

$$24. \quad \|\mathbf{p}\| = \langle \mathbf{p}, \mathbf{p} \rangle^{1/2} = \sqrt{[p(-1)]^2 + [p(0)]^2 + [p(1)]^2 + [p(2)]^2} = \sqrt{(-2)^2 + 0^2 + 2^2 + 10^2} = 6\sqrt{3}$$

$$d(\mathbf{p}, \mathbf{q}) = \|\mathbf{p} - \mathbf{q}\| = \sqrt{[p(-1) - q(-1)]^2 + [p(0) - q(0)]^2 + [p(1) - q(1)]^2 + [p(2) - q(2)]^2} \\ = \sqrt{(-4)^2 + (-1)^2 + 0^2 + 5^2} = \sqrt{42}$$

$$25. \quad \|\mathbf{u}\| = \langle \mathbf{u}, \mathbf{u} \rangle^{1/2} = \left[\left(\begin{bmatrix} 4 & 0 \\ 3 & 5 \end{bmatrix} \begin{bmatrix} -1 \\ 2 \end{bmatrix} \right) \cdot \left(\begin{bmatrix} 4 & 0 \\ 3 & 5 \end{bmatrix} \begin{bmatrix} -1 \\ 2 \end{bmatrix} \right) \right]^{1/2} = \left(\begin{bmatrix} -4 \\ 7 \end{bmatrix} \cdot \begin{bmatrix} -4 \\ 7 \end{bmatrix} \right)^{1/2} = \sqrt{65}$$

$$d(\mathbf{u}, \mathbf{v}) = \|\mathbf{u} - \mathbf{v}\| = \left[\left(\begin{bmatrix} 4 & 0 \\ 3 & 5 \end{bmatrix} \begin{bmatrix} -3 \\ -3 \end{bmatrix} \right) \cdot \left(\begin{bmatrix} 4 & 0 \\ 3 & 5 \end{bmatrix} \begin{bmatrix} -3 \\ -3 \end{bmatrix} \right) \right]^{1/2} = \left(\begin{bmatrix} -12 \\ -24 \end{bmatrix} \cdot \begin{bmatrix} -12 \\ -24 \end{bmatrix} \right)^{1/2} = 12\sqrt{5}$$

$$26. \quad \|\mathbf{u}\| = \langle \mathbf{u}, \mathbf{u} \rangle^{1/2} = \left[\left(\begin{bmatrix} 1 & 2 \\ -1 & 3 \end{bmatrix} \begin{bmatrix} -1 \\ 2 \end{bmatrix} \right) \cdot \left(\begin{bmatrix} 1 & 2 \\ -1 & 3 \end{bmatrix} \begin{bmatrix} -1 \\ 2 \end{bmatrix} \right) \right]^{1/2} = \left(\begin{bmatrix} 3 \\ 7 \end{bmatrix} \cdot \begin{bmatrix} 3 \\ 7 \end{bmatrix} \right)^{1/2} = \sqrt{58}$$

$$d(\mathbf{u}, \mathbf{v}) = \|\mathbf{u} - \mathbf{v}\| = \left[\left(\begin{bmatrix} 1 & 2 \\ -1 & 3 \end{bmatrix} \begin{bmatrix} -3 \\ -3 \end{bmatrix} \right) \cdot \left(\begin{bmatrix} 1 & 2 \\ -1 & 3 \end{bmatrix} \begin{bmatrix} -3 \\ -3 \end{bmatrix} \right) \right]^{1/2} = \left(\begin{bmatrix} -9 \\ -6 \end{bmatrix} \cdot \begin{bmatrix} -9 \\ -6 \end{bmatrix} \right)^{1/2} = 3\sqrt{13}$$

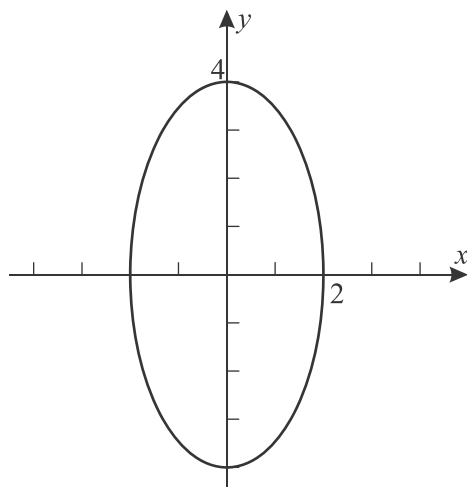
$$27. \quad (a) \quad \langle 2\mathbf{v} - \mathbf{w}, 3\mathbf{u} + 2\mathbf{w} \rangle = \langle 2\mathbf{v}, 3\mathbf{u} + 2\mathbf{w} \rangle - \langle \mathbf{w}, 3\mathbf{u} + 2\mathbf{w} \rangle = \langle 2\mathbf{v}, 3\mathbf{u} \rangle + \langle 2\mathbf{v}, 2\mathbf{w} \rangle - \langle \mathbf{w}, 3\mathbf{u} \rangle - \langle \mathbf{w}, 2\mathbf{w} \rangle \\ = 6\langle \mathbf{v}, \mathbf{u} \rangle + 4\langle \mathbf{v}, \mathbf{w} \rangle - 3\langle \mathbf{w}, \mathbf{u} \rangle - 2\langle \mathbf{w}, \mathbf{w} \rangle = 6\langle \mathbf{u}, \mathbf{v} \rangle + 4\langle \mathbf{v}, \mathbf{w} \rangle - 3\langle \mathbf{u}, \mathbf{w} \rangle - 2\|\mathbf{w}\|^2 \\ = 6(2) + 4(-6) - 3(-3) - 2(49) = -101$$

$$(b) \quad \|\mathbf{u} + \mathbf{v}\| = \sqrt{\langle \mathbf{u} + \mathbf{v}, \mathbf{u} + \mathbf{v} \rangle} = \sqrt{\langle \mathbf{u}, \mathbf{u} + \mathbf{v} \rangle + \langle \mathbf{v}, \mathbf{u} + \mathbf{v} \rangle} = \sqrt{\langle \mathbf{u}, \mathbf{u} \rangle + \langle \mathbf{u}, \mathbf{v} \rangle + \langle \mathbf{v}, \mathbf{u} \rangle + \langle \mathbf{v}, \mathbf{v} \rangle} \\ = \sqrt{\|\mathbf{u}\|^2 + 2\langle \mathbf{u}, \mathbf{v} \rangle + \|\mathbf{v}\|^2} = \sqrt{1 + 2(2) + 4} = 3$$

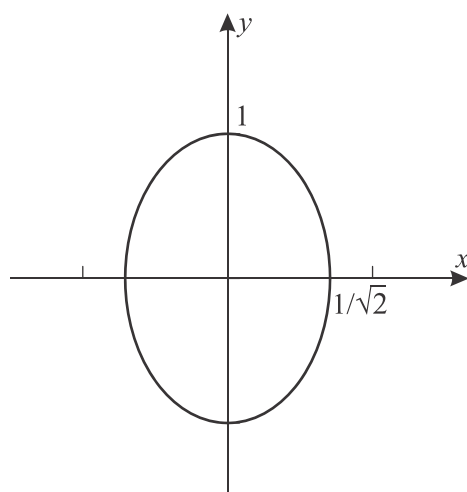
$$28. \quad (a) \quad \langle \mathbf{u} - \mathbf{v} - 2\mathbf{w}, 4\mathbf{u} + \mathbf{v} \rangle = \langle \mathbf{u}, 4\mathbf{u} + \mathbf{v} \rangle - \langle \mathbf{v}, 4\mathbf{u} + \mathbf{v} \rangle - \langle 2\mathbf{w}, 4\mathbf{u} + \mathbf{v} \rangle \\ = \langle \mathbf{u}, 4\mathbf{u} \rangle + \langle \mathbf{u}, \mathbf{v} \rangle - \langle \mathbf{v}, 4\mathbf{u} \rangle - \langle \mathbf{v}, \mathbf{v} \rangle - \langle 2\mathbf{w}, 4\mathbf{u} \rangle - \langle 2\mathbf{w}, \mathbf{v} \rangle \\ = 4\langle \mathbf{u}, \mathbf{u} \rangle + \langle \mathbf{u}, \mathbf{v} \rangle - 4\langle \mathbf{v}, \mathbf{u} \rangle - \langle \mathbf{v}, \mathbf{v} \rangle - 8\langle \mathbf{w}, \mathbf{u} \rangle - 2\langle \mathbf{w}, \mathbf{v} \rangle \\ = 4\|\mathbf{u}\|^2 - 3\langle \mathbf{u}, \mathbf{v} \rangle - \|\mathbf{v}\|^2 - 8\langle \mathbf{u}, \mathbf{w} \rangle - 2\langle \mathbf{v}, \mathbf{w} \rangle = 4 - 3(2) - 4 - 8(-3) - 2(-6) = 30$$

$$(b) \quad \|2\mathbf{w} - \mathbf{v}\| = \sqrt{\langle 2\mathbf{w} - \mathbf{v}, 2\mathbf{w} - \mathbf{v} \rangle} = \sqrt{\langle 2\mathbf{w}, 2\mathbf{w} - \mathbf{v} \rangle - \langle \mathbf{v}, 2\mathbf{w} - \mathbf{v} \rangle} \\ = \sqrt{\langle 2\mathbf{w}, 2\mathbf{w} \rangle - \langle 2\mathbf{w}, \mathbf{v} \rangle - \langle \mathbf{v}, 2\mathbf{w} \rangle + \langle \mathbf{v}, \mathbf{v} \rangle} = \sqrt{4\|\mathbf{w}\|^2 - 4\langle \mathbf{v}, \mathbf{w} \rangle + \|\mathbf{v}\|^2} = \sqrt{4(49) - 4(-6) + 4} \\ = \sqrt{224} = 4\sqrt{14}$$

29. If $\mathbf{u} = (x, y)$ then $\|\mathbf{u}\| = \langle \mathbf{u}, \mathbf{u} \rangle^{1/2} = \sqrt{\frac{1}{4}x^2 + \frac{1}{16}y^2}$, so the equation of the unit circle is $\frac{x^2}{4} + \frac{y^2}{16} = 1$.



30. If $\mathbf{u} = (x, y)$ then $\|\mathbf{u}\| = \langle \mathbf{u}, \mathbf{u} \rangle^{1/2} = \sqrt{2x^2 + y^2}$, so the equation of the unit circle is $2x^2 + y^2 = 1$, which can be rewritten as $\frac{x^2}{1/2} + \frac{y^2}{1} = 1$.



31. $\langle \mathbf{u}, \mathbf{v} \rangle = \frac{1}{9}u_1v_1 + u_2v_2$ (see Example 3)
32. $\langle \mathbf{u}, \mathbf{v} \rangle = \frac{16}{9}u_1v_1 + u_2v_2$ (see Example 3)
33. Axiom 2 does not hold, e.g., with $\mathbf{u} = \mathbf{v} = \mathbf{w} = (1, 0, 0)$ we have $\langle \mathbf{u} + \mathbf{v}, \mathbf{w} \rangle = 4$ but $\langle \mathbf{u}, \mathbf{w} \rangle + \langle \mathbf{v}, \mathbf{w} \rangle = 1 + 1 = 2$;
 Axiom 3 does not hold either, e.g., with $\mathbf{u} = \mathbf{v} = (1, 0, 0)$ and $k = 2$, $\langle k\mathbf{u}, \mathbf{v} \rangle = 4$ does not equal $k\langle \mathbf{u}, \mathbf{v} \rangle = 2$;
 This is not an inner product on R^3 .
34. Axiom 4 does not hold, e.g., $\langle (0, 1, 0), (0, 1, 0) \rangle = -1 < 0$; this is not an inner product on R^3 .

35. By Definition 1, Definition 2, and Theorem 6.1.2, we have

$$\begin{aligned}
 \langle 2\mathbf{v} - 4\mathbf{u}, \mathbf{u} - 3\mathbf{v} \rangle &= \langle 2\mathbf{v} - 4\mathbf{u}, \mathbf{u} \rangle - \langle 2\mathbf{v} - 4\mathbf{u}, 3\mathbf{v} \rangle \\
 &= \langle 2\mathbf{v}, \mathbf{u} \rangle - \langle 4\mathbf{u}, \mathbf{u} \rangle - \langle 2\mathbf{v}, 3\mathbf{v} \rangle + \langle 4\mathbf{u}, 3\mathbf{v} \rangle \\
 &= 2\langle \mathbf{v}, \mathbf{u} \rangle - 4\langle \mathbf{u}, \mathbf{u} \rangle - 6\langle \mathbf{v}, \mathbf{v} \rangle + 12\langle \mathbf{u}, \mathbf{v} \rangle \\
 &= 2\langle \mathbf{u}, \mathbf{v} \rangle - 4\langle \mathbf{u}, \mathbf{u} \rangle - 6\langle \mathbf{v}, \mathbf{v} \rangle + 12\langle \mathbf{u}, \mathbf{v} \rangle \\
 &= 14\langle \mathbf{u}, \mathbf{v} \rangle - 4\|\mathbf{u}\|^2 - 6\|\mathbf{v}\|^2.
 \end{aligned}$$

36. By Definition 1, Definition 2, and Theorem 6.1.2, we have

$$\begin{aligned}
 \langle 5\mathbf{u} + 6\mathbf{v}, 4\mathbf{v} - 3\mathbf{u} \rangle &= \langle 5\mathbf{u} + 6\mathbf{v}, 4\mathbf{v} \rangle - \langle 5\mathbf{u} + 6\mathbf{v}, 3\mathbf{u} \rangle \\
 &= \langle 5\mathbf{u}, 4\mathbf{v} \rangle + \langle 6\mathbf{v}, 4\mathbf{v} \rangle - \langle 5\mathbf{u}, 3\mathbf{u} \rangle - \langle 6\mathbf{v}, 3\mathbf{u} \rangle \\
 &= 20\langle \mathbf{u}, \mathbf{v} \rangle + 24\langle \mathbf{v}, \mathbf{v} \rangle - 15\langle \mathbf{u}, \mathbf{u} \rangle - 18\langle \mathbf{v}, \mathbf{u} \rangle \\
 &= 20\langle \mathbf{u}, \mathbf{v} \rangle + 24\langle \mathbf{v}, \mathbf{v} \rangle - 15\langle \mathbf{u}, \mathbf{u} \rangle - 18\langle \mathbf{u}, \mathbf{v} \rangle \\
 &= 2\langle \mathbf{u}, \mathbf{v} \rangle - 15\|\mathbf{u}\|^2 + 24\|\mathbf{v}\|^2.
 \end{aligned}$$

37. (a) $\langle \mathbf{p}, \mathbf{q} \rangle = \left(\int_{-1}^1 x^2 dx \right) = \left(\frac{x^3}{3} \right)_{-1}^1 = \frac{2}{3}$

(b) $d(\mathbf{p}, \mathbf{q}) = \|\mathbf{p} - \mathbf{q}\| = \left(\int_{-1}^1 (1 - x^2)^2 dx \right)^{1/2} = \left(\left(x - \frac{2x^3}{3} + \frac{x^5}{5} \right)_{-1}^1 \right)^{1/2} = \sqrt{\frac{16}{15}} = \frac{4}{\sqrt{15}}$

(c) $\|\mathbf{p}\| = \langle \mathbf{p}, \mathbf{p} \rangle^{1/2} = \left(\int_{-1}^1 1 dx \right)^{1/2} = \left(x \right)_{-1}^1 = \sqrt{2}$

(d) $\|\mathbf{q}\| = \langle \mathbf{q}, \mathbf{q} \rangle^{1/2} = \left(\int_{-1}^1 x^4 dx \right)^{1/2} = \left(\frac{x^5}{5} \right)_{-1}^1 = \sqrt{\frac{2}{5}}$

38. (a) $\langle \mathbf{p}, \mathbf{q} \rangle = \left(\int_{-1}^1 2x^3(1 - x^3) dx \right) = \left(\left(\frac{x^4}{2} - \frac{2x^7}{7} \right)_{-1}^1 \right) = \frac{-4}{7}$

(b) $d(\mathbf{p}, \mathbf{q}) = \|\mathbf{p} - \mathbf{q}\| = \left(\int_{-1}^1 (3x^3 - 1)^2 dx \right)^{1/2} = \left(\left(\frac{9x^7}{7} - \frac{3x^4}{2} + x \right)_{-1}^1 \right)^{1/2} = \sqrt{\frac{32}{7}} = \frac{4\sqrt{2}}{\sqrt{7}}$

(c) $\|\mathbf{p}\| = \langle \mathbf{p}, \mathbf{p} \rangle^{1/2} = \left(\int_{-1}^1 (2x^3)^2 dx \right)^{1/2} = \left(\frac{4x^7}{7} \right)_{-1}^1 = \sqrt{\frac{8}{7}} = \frac{2\sqrt{2}}{\sqrt{7}}$

(d) $\|\mathbf{q}\| = \langle \mathbf{q}, \mathbf{q} \rangle^{1/2} = \left(\int_{-1}^1 (1 - x^3)^2 dx \right)^{1/2} = \left(\left(x - \frac{x^4}{2} + \frac{x^7}{7} \right)_{-1}^1 \right)^{1/2} = \sqrt{\frac{16}{7}} = \frac{4}{\sqrt{7}}$

39. $\langle \mathbf{f}, \mathbf{g} \rangle = \int_0^1 \cos 2\pi x \sin 2\pi x dx = \frac{1}{2\pi} \frac{(\sin 2\pi x)^2}{2} \Big|_0^1 = 0$ (substituted $u = \sin 2\pi x$)

40. $\langle \mathbf{f}, \mathbf{g} \rangle = \int_0^1 x e^x dx = \left(x e^x - e^x \right) \Big|_0^1 = 1$ (used integration by parts with $u = x$ and $dv = e^x dx$)

41. Part (a) follows directly from Definition 2 and Axiom 4 of Definition 1.

To prove part (b), write

$$\|k\mathbf{v}\| \stackrel{\text{Def.2}}{=} \sqrt{(k\mathbf{v}, k\mathbf{v})} \stackrel{\text{Axiom 3}}{=} \sqrt{k(\mathbf{v}, k\mathbf{v})} \stackrel{\text{Axiom 1}}{=} \sqrt{k(k\mathbf{v}, \mathbf{v})} \stackrel{\text{Axiom 3}}{=} \sqrt{k^2(\mathbf{v}, \mathbf{v})} \stackrel{\text{Def.2}}{=} |k| \|\mathbf{v}\|$$

42. Part (c) follows from Definition 2 and part (b) of Theorem 6.1.1, since

$$d(\mathbf{u}, \mathbf{v}) = \|\mathbf{u} - \mathbf{v}\| = \|(-1)(\mathbf{v} - \mathbf{u})\| = |-1| \|\mathbf{v} - \mathbf{u}\| = \|\mathbf{v} - \mathbf{u}\| = d(\mathbf{v}, \mathbf{u})$$

Part (d) follows from Definition 2 and Axiom 4 of Definition 1.

43. (b) k_1 and k_2 must both be positive in order for $\langle \mathbf{u}, \mathbf{v} \rangle$ to satisfy the positivity axiom. (Refer to the discussion following Theorem 6.1.1.)

44. By using Definition 2 and Axioms 1, 2, and 3 of Definition 1, we have

$$\begin{aligned} \langle \mathbf{u}, \mathbf{v} \rangle &= \left\langle \frac{1}{2}(\mathbf{u} + \mathbf{v}) + \frac{1}{2}(\mathbf{u} - \mathbf{v}), \frac{1}{2}(\mathbf{u} + \mathbf{v}) - \frac{1}{2}(\mathbf{u} - \mathbf{v}) \right\rangle \\ &= \frac{1}{4} \langle \mathbf{u} + \mathbf{v}, \mathbf{u} + \mathbf{v} \rangle + \frac{1}{4} \langle \mathbf{u} - \mathbf{v}, \mathbf{u} + \mathbf{v} \rangle - \frac{1}{4} \langle \mathbf{u} + \mathbf{v}, \mathbf{u} - \mathbf{v} \rangle - \frac{1}{4} \langle \mathbf{u} - \mathbf{v}, \mathbf{u} - \mathbf{v} \rangle \\ &= \frac{1}{4} \|\mathbf{u} + \mathbf{v}\|^2 - \frac{1}{4} \|\mathbf{u} - \mathbf{v}\|^2. \end{aligned}$$

45. By using Definition 2 and Axioms 1, 2, and 3 of Definition 1, we have

$$\begin{aligned} \|\mathbf{u} + \mathbf{v}\|^2 + \|\mathbf{u} - \mathbf{v}\|^2 &= \langle \mathbf{u} + \mathbf{v}, \mathbf{u} + \mathbf{v} \rangle + \langle \mathbf{u} - \mathbf{v}, \mathbf{u} - \mathbf{v} \rangle \\ &= \langle \mathbf{u}, \mathbf{u} \rangle + \langle \mathbf{u}, \mathbf{v} \rangle + \langle \mathbf{v}, \mathbf{u} \rangle + \langle \mathbf{v}, \mathbf{v} \rangle + \langle \mathbf{u}, \mathbf{u} \rangle - \langle \mathbf{u}, \mathbf{v} \rangle - \langle \mathbf{v}, \mathbf{u} \rangle + \langle \mathbf{v}, \mathbf{v} \rangle \\ &= 2\langle \mathbf{u}, \mathbf{u} \rangle + 2\langle \mathbf{v}, \mathbf{v} \rangle \\ &= 2\|\mathbf{u}\|^2 + 2\|\mathbf{v}\|^2. \end{aligned}$$

48. (b) $T(1, 1, 1) = \langle (1, 1, 1), (1, 0, 2) \rangle = (1)(1) + (1)(0) + (1)(2) = 3$

(c) $T(x + x^2) = \langle x + x^2, 1 + x \rangle = (0)(1) + (1)(1) + (1)(0) = 1$

(d) $T(x + x^2) = \langle x + x^2, 1 + x \rangle = (1 + 1^2)(1 + 1) + (0 + 0^2)(1 + 0) + (-1 + (-1)^2)(1 - 1) = 4$

True-False Exercises

(a) True. The dot product is the special case of the weighted inner product with all the weights equal to 1.

(b) False. For example, if $\langle \mathbf{u}, \mathbf{v} \rangle = \mathbf{u} \cdot \mathbf{v}$, $\mathbf{u} = (1, 1)$, and $\mathbf{v} = (-2, 1)$ then $\langle \mathbf{u}, \mathbf{v} \rangle = -1$.

(c) True. This follows from Axioms 1 and 2 of Definition 1 since $\langle \mathbf{u}, \mathbf{v} + \mathbf{w} \rangle = \langle \mathbf{v} + \mathbf{w}, \mathbf{u} \rangle = \langle \mathbf{v}, \mathbf{u} \rangle + \langle \mathbf{w}, \mathbf{u} \rangle$.

(d) True. This follows from Axiom 3 of Definition 1 as well as part (e) of Theorem 6.1.2.

(e) False. For example, if $\langle \mathbf{u}, \mathbf{v} \rangle = \mathbf{u} \cdot \mathbf{v}$, $\mathbf{u} = (1, 1)$, and $\mathbf{v} = (-1, 1)$ then $\langle \mathbf{u}, \mathbf{v} \rangle = 0$ even though both vectors are nonzero.

(f) True. By Definition 2, $\|\mathbf{v}\|^2 = \langle \mathbf{v}, \mathbf{v} \rangle$ so by Axiom 4 of Definition 1, $\|\mathbf{v}\|^2 = 0$ implies $\mathbf{v} = \mathbf{0}$.

(g) False. A must be invertible; otherwise $A\mathbf{v} = \mathbf{0}$ has nontrivial solutions $\mathbf{v} \neq \mathbf{0}$ even though $\langle \mathbf{v}, \mathbf{v} \rangle = A\mathbf{v} \cdot A\mathbf{v} = \mathbf{0}$ which would violate Axiom 4 of Definition 1.

6.2 Angle and Orthogonality in Inner Product Spaces

1. (a) $\cos \theta = \frac{\langle \mathbf{u}, \mathbf{v} \rangle}{\|\mathbf{u}\| \|\mathbf{v}\|} = \frac{(1)(2) + (-3)(4)}{\sqrt{1^2 + (-3)^2} \sqrt{2^2 + 4^2}} = -\frac{10}{\sqrt{10} \sqrt{20}} = -\frac{10}{\sqrt{200}} = -\frac{10}{10\sqrt{2}} = -\frac{1}{\sqrt{2}}$
 (b) $\cos \theta = \frac{\langle \mathbf{u}, \mathbf{v} \rangle}{\|\mathbf{u}\| \|\mathbf{v}\|} = \frac{(-1)(2) + (5)(4) + (2)(-9)}{\sqrt{(-1)^2 + 5^2 + 2^2} \sqrt{2^2 + 4^2 + (-9)^2}} = 0$
 (c) $\cos \theta = \frac{\langle \mathbf{u}, \mathbf{v} \rangle}{\|\mathbf{u}\| \|\mathbf{v}\|} = \frac{(1)(-3) + (0)(-3) + (1)(-3) + (0)(-3)}{\sqrt{1^2 + 0^2 + 1^2 + 0^2} \sqrt{(-3)^2 + (-3)^2 + (-3)^2 + (-3)^2}} = -\frac{6}{\sqrt{2} \sqrt{36}} = -\frac{1}{\sqrt{2}}$
2. (a) $\cos \theta = \frac{\langle \mathbf{u}, \mathbf{v} \rangle}{\|\mathbf{u}\| \|\mathbf{v}\|} = \frac{(-1)(3) + (0)(8)}{\sqrt{(-1)^2 + 0^2} \sqrt{3^2 + 8^2}} = -\frac{3}{\sqrt{73}}$
 (b) $\cos \theta = \frac{\langle \mathbf{u}, \mathbf{v} \rangle}{\|\mathbf{u}\| \|\mathbf{v}\|} = \frac{(4)(1) + (1)(0) + (8)(-3)}{\sqrt{4^2 + 1^2 + 8^2} \sqrt{1^2 + 0^2 + (-3)^2}} = -\frac{20}{\sqrt{81} \sqrt{10}} = -\frac{20}{9\sqrt{10}}$
 (c) $\cos \theta = \frac{\langle \mathbf{u}, \mathbf{v} \rangle}{\|\mathbf{u}\| \|\mathbf{v}\|} = \frac{(2)(4) + (1)(0) + (7)(0) + (-1)(0)}{\sqrt{2^2 + 1^2 + 7^2 + (-1)^2} \sqrt{4^2 + 0^2 + 0^2 + 0^2}} = \frac{8}{4\sqrt{55}} = \frac{2}{\sqrt{55}}$
3. $\cos \theta = \frac{\langle \mathbf{p}, \mathbf{q} \rangle}{\|\mathbf{p}\| \|\mathbf{q}\|} = \frac{(-1)(2) + (5)(4) + (2)(-9)}{\sqrt{(-1)^2 + 5^2 + 2^2} \sqrt{2^2 + 4^2 + (-9)^2}} = 0$
4. $\cos \theta = \frac{\langle \mathbf{p}, \mathbf{q} \rangle}{\|\mathbf{p}\| \|\mathbf{q}\|} = \frac{(0)(7) + (1)(3) + (-1)(3)}{\sqrt{0^2 + 1^2 + (-1)^2} \sqrt{7^2 + 3^2 + 3^2}} = 0$
5. $\cos \theta = \frac{\langle \mathbf{U}, \mathbf{V} \rangle}{\|\mathbf{U}\| \|\mathbf{V}\|} = \frac{\text{tr}(\mathbf{U}^T \mathbf{V})}{\sqrt{\text{tr}(\mathbf{U}^T \mathbf{U})} \sqrt{\text{tr}(\mathbf{V}^T \mathbf{V})}} = \frac{(2)(3) + (6)(2) + (1)(1) + (-3)(0)}{\sqrt{2^2 + 6^2 + 1^2 + (-3)^2} \sqrt{3^2 + 2^2 + 1^2 + 0^2}} = \frac{19}{\sqrt{50} \sqrt{14}} = \frac{19}{10\sqrt{7}}$
6. $\cos \theta = \frac{\langle \mathbf{U}, \mathbf{V} \rangle}{\|\mathbf{U}\| \|\mathbf{V}\|} = \frac{\text{tr}(\mathbf{U}^T \mathbf{V})}{\sqrt{\text{tr}(\mathbf{U}^T \mathbf{U})} \sqrt{\text{tr}(\mathbf{V}^T \mathbf{V})}} = \frac{(2)(-3) + (4)(1) + (-1)(4) + (3)(2)}{\sqrt{2^2 + 4^2 + (-1)^2 + 3^2} \sqrt{(-3)^2 + 1^2 + 4^2 + 2^2}} = 0$
7. (a) orthogonal: $\langle \mathbf{u}, \mathbf{v} \rangle = -4 + 6 - 2 = 0$
 (b) not orthogonal: $\langle \mathbf{u}, \mathbf{v} \rangle = -2 - 2 - 2 = -6 \neq 0$
 (c) orthogonal: $\langle \mathbf{u}, \mathbf{v} \rangle = (a)(-b) + (b)(a) = 0$
8. (a) orthogonal: $\langle \mathbf{u}, \mathbf{v} \rangle = 0 + 0 + 0 = 0$
 (b) not orthogonal: $\langle \mathbf{u}, \mathbf{v} \rangle = -8 + 6 + 20 + 9 = 27 \neq 0$
 (c) orthogonal: $\langle \mathbf{u}, \mathbf{v} \rangle = -ac + 0 + ac = 0$
9. $\langle \mathbf{p}, \mathbf{q} \rangle = (-1)(0) + (-1)(2) + (2)(1) = 0$
10. $\langle \mathbf{p}, \mathbf{q} \rangle = (2)(4) + (-3)(2) + (1)(-2) = 0$
11. $\langle \mathbf{U}, \mathbf{V} \rangle = (2)(-3) + (1)(0) + (-1)(0) + (3)(2) = 0$
12. $\langle \mathbf{U}, \mathbf{V} \rangle = (5)(1) + (-1)(3) + (2)(-1) + (-2)(0) = 0$

13. The vectors are not orthogonal with respect to the Euclidean inner product since $\langle \mathbf{u}, \mathbf{v} \rangle = (1)(2) + (3)(-1) = -1 \neq 0$. Using the weighted inner product instead yields $\langle \mathbf{u}, \mathbf{v} \rangle = 2(1)(2) + k(3)(-1) = 4 - 3k$, so the vectors are orthogonal with respect to this inner product if $k = \frac{4}{3}$.
14. The vectors are not orthogonal with respect to the Euclidean inner product since $\langle \mathbf{u}, \mathbf{v} \rangle = (2)(0) + (-4)(3) = -12 \neq 0$. Using the weighted inner product instead yields $\langle \mathbf{u}, \mathbf{v} \rangle = 2(2)(0) + k(-4)(3) = -12k$. This would equal 0 if $k = 0$, however, the corresponding formula $\langle \mathbf{u}, \mathbf{v} \rangle = 2u_1v_1$ does not represent an inner product since it violates Axiom 4. Consequently, no k values can be found for which the vectors are orthogonal.
15. The orthogonality of the two vectors implies $(w_1)(1)(2) + (w_2)(2)(-4) = 0$. The weights must be positive numbers such that $w_1 = 4w_2$.
16. Begin by forming a matrix A whose rows are the given vectors $\mathbf{u}$, $\mathbf{v}$, and $\mathbf{w}$: $A = \begin{bmatrix} 2 & 1 & -4 & 0 \\ -1 & -1 & 2 & 2 \\ 3 & 2 & 5 & 4 \end{bmatrix}$. The reduced row echelon form of A is $\begin{bmatrix} 1 & 0 & 0 & \frac{34}{11} \\ 0 & 1 & 0 & -4 \\ 0 & 0 & 1 & \frac{6}{11} \end{bmatrix}$ therefore the general solution of the homogeneous system $A\mathbf{x} = \mathbf{0}$ is $x_1 = -\frac{34}{11}t$, $x_2 = 4t$, $x_3 = -\frac{6}{11}t$, $x_4 = t$, so that all solution vectors are scalar multiples of $(-34, 44, -6, 11)$. All solution vectors of the homogeneous system are orthogonal to every row vector of the coefficient matrix (see Example 6 in Section 6.2).
The magnitude of the vector $(-34, 44, -6, 11)$ is $\sqrt{(-34)^2 + 44^2 + (-6)^2 + 11^2} = 57$.
We conclude that the two unit vectors that are orthogonal to all three of the vectors $\mathbf{u}$, $\mathbf{v}$, and $\mathbf{w}$ are $\frac{1}{57}(-34, 44, -6, 11) = (-\frac{34}{57}, \frac{44}{57}, -\frac{6}{57}, \frac{11}{57})$ and $-\frac{1}{57}(-34, 44, -6, 11) = (\frac{34}{57}, -\frac{44}{57}, \frac{6}{57}, -\frac{11}{57})$.
17. Orthogonality of $\mathbf{p}_1$ and $\mathbf{p}_3$ implies $\langle \mathbf{p}_1, \mathbf{p}_3 \rangle = (2)(1) + (k)(2) + (6)(3) = 2k + 20 = 0$ so $k = -10$. Likewise, orthogonality of $\mathbf{p}_2$ and $\mathbf{p}_3$ implies $\langle \mathbf{p}_2, \mathbf{p}_3 \rangle = (l)(1) + (5)(2) + (3)(3) = l + 19$ so $l = -19$.
Substituting the values of k and l obtained above yields the polynomials $\mathbf{p}_1 = 2 - 10x + 6x^2$ and $\mathbf{p}_2 = -19 + 5x + 3x^2$ which are not orthogonal since $\langle \mathbf{p}_1, \mathbf{p}_2 \rangle = (2)(-19) + (-10)(5) + (6)(3) = -70 \neq 0$. We conclude that no scalars k and l exist that make the three vectors mutually orthogonal.
18. $\langle \mathbf{u}, \mathbf{v} \rangle = \left(\begin{bmatrix} 2 & 1 \\ 1 & 1 \end{bmatrix} \begin{bmatrix} 3 \\ 3 \end{bmatrix} \right) \cdot \left(\begin{bmatrix} 2 & 1 \\ 1 & 1 \end{bmatrix} \begin{bmatrix} 5 \\ -8 \end{bmatrix} \right) = \begin{bmatrix} 9 \\ 6 \end{bmatrix} \cdot \begin{bmatrix} 2 \\ -3 \end{bmatrix} = 0$
19. $\langle \mathbf{p}, \mathbf{q} \rangle = p(-2)q(-2) + p(0)q(0) + p(2)q(2) = (-2)(4) + (0)(0) + (2)(4) = 0$
20. For A to be a in the subspace of M_{22} spanned by U and V , there must exist scalars a and b such that

$$a \begin{bmatrix} 1 & -1 \\ 3 & 0 \end{bmatrix} + b \begin{bmatrix} 4 & 0 \\ 9 & 2 \end{bmatrix} = \begin{bmatrix} -1 & 1 \\ 0 & 2 \end{bmatrix}.$$

Equating corresponding entries in the second column on both sides yields $a = -1$ and $b = 1$. However, for these values, the remaining entries on both sides do not equal (e.g., $(-1)(1) + (1)(4) \neq -1$). We conclude that A is not in the subspace of M_{22} spanned by U and V .

21. $|\langle \mathbf{u}, \mathbf{v} \rangle| = |2(1)(2) + 3(0)(1) + (3)(-1)| = |1| = 1;$

$$\|\mathbf{u}\| = \sqrt{\langle \mathbf{u}, \mathbf{u} \rangle} = \sqrt{2(1)(1) + 3(0)(0) + (3)(3)} = \sqrt{11};$$

$$\|\mathbf{v}\| = \sqrt{\langle \mathbf{v}, \mathbf{v} \rangle} = \sqrt{2(2)(2) + 3(1)(1) + (-1)(-1)} = \sqrt{12};$$

since $\|\mathbf{u}\| \|\mathbf{v}\| = \sqrt{132} \geq 1 = |\langle \mathbf{u}, \mathbf{v} \rangle|$, we conclude that the Cauchy-Schwarz inequality holds.

22. $|\langle U, V \rangle| = |(-1)(1) + (2)(0) + (6)(3) + (1)(3)| = |20| = 20;$

$$\|U\| = \sqrt{\langle U, U \rangle} = \sqrt{(-1)(-1) + (2)(2) + (6)(6) + (1)(1)} = \sqrt{42};$$

$$\|V\| = \sqrt{\langle V, V \rangle} = \sqrt{(1)(1) + (0)(0) + (3)(3) + (3)(3)} = \sqrt{19};$$

since $\|U\| \|V\| = \sqrt{798} \geq \sqrt{400} = 20 = |\langle U, V \rangle|$, we conclude that the Cauchy-Schwarz inequality holds.

23. $|\langle \mathbf{p}, \mathbf{q} \rangle| = |(-1)(2) + (2)(0) + (1)(-4)| = |-6| = 6;$

$$\|\mathbf{p}\| = \sqrt{\langle \mathbf{p}, \mathbf{p} \rangle} = \sqrt{(-1)(-1) + (2)(2) + (1)(1)} = \sqrt{6};$$

$$\|\mathbf{q}\| = \sqrt{\langle \mathbf{q}, \mathbf{q} \rangle} = \sqrt{(2)(2) + (0)(0) + (-4)(-4)} = \sqrt{20};$$

since $\|\mathbf{p}\| \|\mathbf{q}\| = \sqrt{120} \geq \sqrt{36} = 6 = |\langle \mathbf{p}, \mathbf{q} \rangle|$, we conclude that the Cauchy-Schwarz inequality holds.

24. $|\langle \mathbf{u}, \mathbf{v} \rangle| = \left| \left(\begin{bmatrix} 2 & 1 \\ 1 & 1 \end{bmatrix} \begin{bmatrix} 1 \\ 1 \end{bmatrix} \right) \cdot \left(\begin{bmatrix} 2 & 1 \\ 1 & 1 \end{bmatrix} \begin{bmatrix} 1 \\ -1 \end{bmatrix} \right) \right| = \left| \begin{bmatrix} 3 \\ 2 \end{bmatrix} \cdot \begin{bmatrix} 1 \\ 0 \end{bmatrix} \right| = |3| = 3;$

$$\|\mathbf{u}\| = \sqrt{\langle \mathbf{u}, \mathbf{u} \rangle} = \sqrt{\left(\begin{bmatrix} 2 & 1 \\ 1 & 1 \end{bmatrix} \begin{bmatrix} 1 \\ 1 \end{bmatrix} \right) \cdot \left(\begin{bmatrix} 2 & 1 \\ 1 & 1 \end{bmatrix} \begin{bmatrix} 1 \\ 1 \end{bmatrix} \right)} = \sqrt{\begin{bmatrix} 3 \\ 2 \end{bmatrix} \cdot \begin{bmatrix} 3 \\ 2 \end{bmatrix}} = \sqrt{13};$$

$$\|\mathbf{v}\| = \sqrt{\langle \mathbf{v}, \mathbf{v} \rangle} = \sqrt{\left(\begin{bmatrix} 2 & 1 \\ 1 & 1 \end{bmatrix} \begin{bmatrix} 1 \\ -1 \end{bmatrix} \right) \cdot \left(\begin{bmatrix} 2 & 1 \\ 1 & 1 \end{bmatrix} \begin{bmatrix} 1 \\ -1 \end{bmatrix} \right)} = \sqrt{\begin{bmatrix} 1 \\ 0 \end{bmatrix} \cdot \begin{bmatrix} 1 \\ 0 \end{bmatrix}} = \sqrt{1} = 1;$$

since $\|\mathbf{u}\| \|\mathbf{v}\| = \sqrt{13} \geq \sqrt{9} = 3 = |\langle \mathbf{u}, \mathbf{v} \rangle|$, we conclude that the Cauchy-Schwarz inequality holds.

25. By inspection, $\langle \mathbf{u}, \mathbf{w}_1 \rangle = -2 \neq 0$. Since $\mathbf{u}$ is not orthogonal to $\mathbf{w}_1$, it is not orthogonal to the subspace.

26. We have $\langle \mathbf{p}, \mathbf{w}_1 \rangle = (-1)(2) + (-1)(0) + (2)(-1) + (4)(1) = 0$ and

$$\langle \mathbf{p}, \mathbf{w}_2 \rangle = (-1)(0) + (-1)(4) + (2)(-2) + (4)(2) = 0 \text{ therefore for all scalars } a \text{ and } b$$

$$\langle \mathbf{p}, a\mathbf{w}_1 + b\mathbf{w}_2 \rangle = a\langle \mathbf{p}, \mathbf{w}_1 \rangle + b\langle \mathbf{p}, \mathbf{w}_2 \rangle = 0. \text{ We conclude that } \mathbf{p} \text{ is orthogonal to } \text{span}\{\mathbf{w}_1, \mathbf{w}_2\}.$$

27. Begin by forming a matrix A whose rows are the given vectors:

$$A = \begin{bmatrix} 1 & 4 & 5 & 2 \\ 2 & 1 & 3 & 0 \\ -1 & 3 & 2 & 2 \end{bmatrix} \text{ has the reduced row echelon form } \begin{bmatrix} 1 & 0 & 1 & -\frac{2}{7} \\ 0 & 1 & 1 & \frac{4}{7} \\ 0 & 0 & 0 & 0 \end{bmatrix}. \text{ The general solution of the}$$

homogeneous system $A\mathbf{x} = \mathbf{0}$ is $x_1 = -s + \frac{2}{7}t$, $x_2 = -s - \frac{4}{7}t$, $x_3 = s$, $x_4 = t$, therefore

$$\mathbf{x} = s(-1, -1, 1, 0) + t\left(\frac{2}{7}, -\frac{4}{7}, 0, 1\right).$$

A basis for the orthogonal complement is formed by vectors $(-1, -1, 1, 0)$ and $(\frac{2}{7}, -\frac{4}{7}, 0, 1)$.

28. Begin by forming a matrix A whose rows are the given vectors:

$$A = \begin{bmatrix} 1 & 4 & 5 & 6 & 9 \\ 3 & -2 & 1 & 4 & -1 \\ -1 & 0 & -1 & -2 & -1 \\ 2 & 3 & 5 & 7 & 8 \end{bmatrix} \text{ has the reduced row echelon form } \begin{bmatrix} 1 & 0 & 1 & 2 & 1 \\ 0 & 1 & 1 & 1 & 2 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix}. \text{ The general solution of}$$

the homogeneous system $A\mathbf{x} = \mathbf{0}$ is $x_1 = -r - 2s - t$, $x_2 = -r - s - 2t$, $x_3 = r$, $x_4 = s$, $x_5 = t$, therefore

$$\mathbf{x} = r(-1, -1, 1, 0, 0) + s(-2, -1, 0, 1, 0) + t(-1, -2, 0, 0, 1).$$

A basis for the orthogonal complement is formed by vectors $(-1, -1, 1, 0, 0)$, $(-2, -1, 0, 1, 0)$, and $(-1, -2, 0, 0, 1)$.

29. (a) Every vector in W has form $(x, y) = (x, 2x)$, i.e., $W = \text{span}\{(1, 2)\}$. By inspection, all vectors in R^2 orthogonal to $(1, 2)$ are scalar multiples of the vector $(2, -1)$. Eliminating t from

$$(x, y) = t(2, -1) = (2t, -t) \text{ we obtain } x = 2(-y), \text{ i.e. } W^\perp \text{ can be represented using an equation}$$

$$y = -\frac{1}{2}x.$$

(An alternate method of solving this exercise is to follow the procedure of Example 6: letting

$$A = \begin{bmatrix} 1 & 2 \end{bmatrix}, \text{ the general solution of } A \begin{bmatrix} x \\ y \end{bmatrix} = 0 \text{ is } x = -2t, y = t. \text{ Eliminating } t \text{ yields } y = -\frac{1}{2}x.)$$

- (b) $W^\perp$ will have dimension 1. A normal to the plane is $\mathbf{u} = (1, -2, -3)$, so $W^\perp$ will consist of all scalar multiples of $\mathbf{u}$ or $t\mathbf{u} = (t, -2t, -3t)$ so parametric equations for $W^\perp$ are $x = t$,

$$y = -2t, z = -3t.$$

30. (a) Every vector in W has form $(0, y, 0)$, i.e. $W = \text{span}\{(0, 1, 0)\}$. By inspection, all vectors in R^3 orthogonal to $(0, 1, 0)$ are linear combinations of $(1, 0, 0)$ and $(0, 0, 1)$ i.e. $W^\perp$ is the xz plane.

- (b) Every vector in W has form $(0, y, z)$, i.e. $W = \text{span}\{(0, 1, 0), (0, 0, 1)\}$. By inspection, all vectors in R^3 orthogonal to W have form $(1, 0, 0)$ i.e. $W^\perp$ is the x axis.

31. (a) $\langle \mathbf{p}, \mathbf{q} \rangle = \int_0^1 x^3 dx = \left[\frac{x^4}{4} \right]_0^1 = \frac{1}{4}$

$$(b) \quad \|\mathbf{p}\| = \langle \mathbf{p}, \mathbf{p} \rangle^{\frac{1}{2}} = \left(\int_0^1 x^2 \, dx \right)^{\frac{1}{2}} = \left(\frac{x^3}{3} \Big|_0^1 \right)^{\frac{1}{2}} = \frac{1}{\sqrt{3}}$$

$$\|\mathbf{q}\| = \langle \mathbf{q}, \mathbf{q} \rangle^{1/2} = \left(\int_0^1 x^4 \, dx \right)^{1/2} = \left(\frac{x^5}{5} \Big|_0^1 \right)^{1/2} = \frac{1}{\sqrt{5}}$$

$$32. (a) \quad \text{Using the results obtained in Exercise 31 we have } \cos \theta = \frac{\langle \mathbf{p}, \mathbf{q} \rangle}{\|\mathbf{p}\| \|\mathbf{q}\|} = \frac{\frac{1}{4}}{\frac{1}{\sqrt{3}} \cdot \frac{1}{\sqrt{5}}} = \frac{\sqrt{15}}{4}.$$

$$(b) \quad d(\mathbf{p}, \mathbf{q}) = \|\mathbf{p} - \mathbf{q}\| = \left(\int_0^1 (x - x^2)^2 \, dx \right)^{1/2} = \left(\left(\frac{x^3}{3} - \frac{x^4}{2} + \frac{x^5}{5} \right) \Big|_0^1 \right)^{1/2} = \sqrt{\frac{1}{30}}$$

$$33. (a) \quad \langle \mathbf{p}, \mathbf{q} \rangle = \int_{-1}^1 (x^2 - x)(x + 1) \, dx = \int_{-1}^1 (x^3 - x) \, dx = \left(\frac{x^4}{4} - \frac{x^2}{2} \right) \Big|_{-1}^1 = 0$$

$$(b) \quad \|\mathbf{p}\| = \langle \mathbf{p}, \mathbf{p} \rangle^{\frac{1}{2}} = \left(\int_{-1}^1 (x^2 - x)^2 \, dx \right)^{\frac{1}{2}} = \left(\left(\frac{x^5}{5} - \frac{x^4}{2} + \frac{x^3}{3} \right) \Big|_{-1}^1 \right)^{\frac{1}{2}} = \frac{4}{\sqrt{15}}$$

$$\|\mathbf{q}\| = \langle \mathbf{q}, \mathbf{q} \rangle^{1/2} = \left(\int_{-1}^1 (x + 1)^2 \, dx \right)^{1/2} = \left(\left(\frac{x^3}{3} + x^2 + x \right) \Big|_{-1}^1 \right)^{1/2} = \sqrt{\frac{8}{3}} = 2\sqrt{\frac{2}{3}}$$

$$34. (a) \quad \text{Using the results obtained in Exercise 33 we have } \cos \theta = \frac{\langle \mathbf{p}, \mathbf{q} \rangle}{\|\mathbf{p}\| \|\mathbf{q}\|} = 0.$$

$$(b) \quad d(\mathbf{p}, \mathbf{q}) = \|\mathbf{p} - \mathbf{q}\| = \left(\int_{-1}^1 (x^2 - 2x - 1)^2 \, dx \right)^{1/2} = \left(\left(\frac{x^5}{5} - x^4 + \frac{2x^3}{3} + 2x^2 + x \right) \Big|_{-1}^1 \right)^{1/2} = 2\sqrt{\frac{14}{15}}$$

$$35. (a) \quad \langle \mathbf{p}, \mathbf{q} \rangle = \int_0^1 \left(\frac{1}{2} - x \right) \, dx = \left(\frac{1}{2}x - \frac{x^2}{2} \right) \Big|_0^1 = 0$$

$$(b) \quad \|\mathbf{p} + \mathbf{q}\|^2 = \int_0^1 \left(\frac{3}{2} - x \right)^2 \, dx = \left(\frac{9}{4}x - \frac{3x^2}{2} + \frac{x^3}{3} \right) \Big|_0^1 = \frac{13}{12}; \quad \|\mathbf{p}\|^2 = \int_0^1 (1)^2 \, dx = x \Big|_0^1 = 1;$$

$$\mathbf{q}^2 = \int_0^1 \left(\frac{1}{2} - x \right)^2 \, dx = \left(\frac{x}{4} - \frac{x^2}{2} + \frac{x^3}{3} \right) \Big|_0^1 = \frac{1}{12}; \text{ we conclude that } \|\mathbf{p} + \mathbf{q}\|^2 = \frac{13}{12} = \|\mathbf{p}\|^2 + \|\mathbf{q}\|^2.$$

$$36. (a) \quad \langle \mathbf{p}, \mathbf{q} \rangle = \int_{-1}^1 (x^3 - x) \, dx = \left(\frac{x^4}{4} - \frac{x^2}{2} \right) \Big|_{-1}^1 = 0$$

$$(b) \quad \|\mathbf{p} + \mathbf{q}\|^2 = \int_{-1}^1 (x^2 + x - 1)^2 \, dx = \left(\frac{x^5}{5} + \frac{x^4}{2} - \frac{x^3}{3} - x^2 + x \right) \Big|_{-1}^1 = \frac{26}{15};$$

$$\|\mathbf{p}\|^2 = \int_{-1}^1 (x)^2 \, dx = \frac{x^3}{3} \Big|_{-1}^1 = \frac{2}{3}; \quad \|\mathbf{q}\|^2 = \int_{-1}^1 (x^2 - 1)^2 \, dx = \left(\frac{x^5}{5} - \frac{2x^3}{3} + x \right) \Big|_{-1}^1 = \frac{16}{15};$$

$$\text{we conclude that } \|\mathbf{p} + \mathbf{q}\|^2 = \frac{26}{15} = \|\mathbf{p}\|^2 + \|\mathbf{q}\|^2.$$

$$37. \quad \|\mathbf{u} - \mathbf{v}\|^2 = \langle \mathbf{u} - \mathbf{v}, \mathbf{u} - \mathbf{v} \rangle = \langle \mathbf{u}, \mathbf{u} - \mathbf{v} \rangle - \langle \mathbf{v}, \mathbf{u} - \mathbf{v} \rangle = \langle \mathbf{u}, \mathbf{u} \rangle - \langle \mathbf{u}, \mathbf{v} \rangle - \langle \mathbf{v}, \mathbf{u} \rangle + \langle \mathbf{v}, \mathbf{v} \rangle = 1 - 0 - 0 + 1 = 2 \text{ therefore } \|\mathbf{u} - \mathbf{v}\| = \sqrt{2}.$$

$$38. \quad \text{Assuming } \langle \mathbf{w}, \mathbf{u}_1 \rangle = \langle \mathbf{w}, \mathbf{u}_2 \rangle = 0, \text{ it follows that}$$

$$\langle \mathbf{w}, k_1 \mathbf{u}_1 + k_2 \mathbf{u}_2 \rangle = \langle \mathbf{w}, k_1 \mathbf{u}_1 \rangle + \langle \mathbf{w}, k_2 \mathbf{u}_2 \rangle = k_1 \langle \mathbf{w}, \mathbf{u}_1 \rangle + k_2 \langle \mathbf{w}, \mathbf{u}_2 \rangle = (k_1)(0) + (k_2)(0) = 0.$$

When V is R^3 with the Euclidean inner product this means that a vector perpendicular to two vectors is also perpendicular to the entire plane spanned by these vectors (assuming they are noncollinear).

39. Using the trigonometric identity $\cos \alpha \cos \beta = \frac{1}{2} \cos(\alpha - \beta) + \frac{1}{2} \cos(\alpha + \beta)$ we obtain

$$\mathbf{f}_k, \mathbf{f}_l = \frac{1}{2} \int_0^\pi \cos((k-l)x) dx + \frac{1}{2} \int_0^\pi \cos((k+l)x) dx \text{ where both } k-l \text{ and } k+l \text{ are nonzero integers.}$$

Substituting $u = (k-l)x$ in the first integral, and $t = (k+l)x$ in the second integral yields

$$\langle \mathbf{f}_k, \mathbf{f}_l \rangle = \frac{1}{2} \frac{\sin((k-l)x)}{k-l} \Big|_0^\pi + \frac{1}{2} \frac{\sin((k+l)x)}{k+l} \Big|_0^\pi = 0 + 0 = 0 \text{ since } \sin(m\pi) = 0 \text{ for any integer } m.$$

40. We are looking for positive real numbers w_1 and w_2 for which the inner product $\langle \mathbf{u}, \mathbf{v} \rangle = w_1 u_1 v_1 + w_2 u_2 v_2$ satisfies $\langle \mathbf{u}, \mathbf{v} \rangle = 0$, $\|\mathbf{u}\|^2 = \langle \mathbf{u}, \mathbf{u} \rangle = 1$, and $\|\mathbf{v}\|^2 = \langle \mathbf{v}, \mathbf{v} \rangle = 1$ when applied to the given two vectors. These three equations yield a linear system

$$\begin{aligned} -w_1 + 3w_2 &= 0 \\ w_1 + 3w_2 &= 1 \\ w_1 + 3w_2 &= 1 \end{aligned}$$

The augmented matrix of this system has the reduced row echelon form $\begin{bmatrix} 1 & 0 & \frac{1}{2} \\ 0 & 1 & \frac{1}{6} \\ 0 & 0 & 0 \end{bmatrix}$. Since the system has

only one solution $w_1 = \frac{1}{2}$, $w_2 = \frac{1}{6}$, we conclude that the weighted Euclidean inner product that satisfies the given conditions is $\langle \mathbf{u}, \mathbf{v} \rangle = \frac{1}{2} u_1 v_1 + \frac{1}{6} u_2 v_2$.

41. $\text{span}\{\mathbf{u}_1, \mathbf{u}_2, \dots, \mathbf{u}_r\}$ contains all linear combinations $k_1 \mathbf{u}_1 + k_2 \mathbf{u}_2 + \dots + k_r \mathbf{u}_r$

where $k_1, k_2, \dots, k_r$ are arbitrary scalars. Let $\mathbf{v} \in \text{span}\{\mathbf{u}_1, \mathbf{u}_2, \dots, \mathbf{u}_r\}$.

$$\langle \mathbf{w}, \mathbf{v} \rangle = \langle \mathbf{w}, k_1 \mathbf{u}_1 + k_2 \mathbf{u}_2 + \dots + k_r \mathbf{u}_r \rangle = k_1 \langle \mathbf{w}, \mathbf{u}_1 \rangle + k_2 \langle \mathbf{w}, \mathbf{u}_2 \rangle + \dots + k_r \langle \mathbf{w}, \mathbf{u}_r \rangle = 0 + 0 + \dots + 0 = 0$$

Thus if $\mathbf{w}$ is orthogonal to each vector $\mathbf{u}_1, \mathbf{u}_2, \dots, \mathbf{u}_r$, then $\mathbf{w}$ must be orthogonal to every vector in $\text{span}\{\mathbf{u}_1, \mathbf{u}_2, \dots, \mathbf{u}_r\}$.

42. Let $\mathbf{w}$ be a vector orthogonal to all of the basis vectors. It must be possible to express $\mathbf{w}$ as a linear combination of the basis vectors: $\mathbf{w} = k_1 \mathbf{v}_1 + k_2 \mathbf{v}_2 + \dots + k_r \mathbf{v}_r$. By the Axioms 1-3 of Definition 1 in Section 6.1, we can write

$$\begin{aligned} \langle \mathbf{w}, \mathbf{w} \rangle &= \langle \mathbf{w}, k_1 \mathbf{v}_1 + k_2 \mathbf{v}_2 + \dots + k_r \mathbf{v}_r \rangle \\ &= k_1 \langle \mathbf{w}, \mathbf{v}_1 \rangle + k_2 \langle \mathbf{w}, \mathbf{v}_2 \rangle + \dots + k_r \langle \mathbf{w}, \mathbf{v}_r \rangle \\ &= k_1 \cdot 0 + k_2 \cdot 0 + \dots + k_r \cdot 0 \\ &= 0 \end{aligned}$$

By Axiom 4 of Definition 1 in Section 6.1, we must have $\mathbf{w} = \mathbf{0}$.

43. Suppose that $\mathbf{v}$ is orthogonal to every basis vector. Then, as in Exercise 41, $\mathbf{v}$ is orthogonal to the span of the set of basis vectors, which is all of W , hence $\mathbf{v}$ is in $W^\perp$. If $\mathbf{v}$ is not orthogonal to every basis vector, then $\mathbf{v}$ clearly cannot be in $W^\perp$. Thus $W^\perp$ consists of all vectors orthogonal to every basis vector.
44. This result can be established by induction.
46. Using the Euclidean inner product, apply the Cauchy-Schwarz inequality to $\mathbf{u} = (a, b)$ and $\mathbf{v} = (\cos \theta, \sin \theta)$.
47. Using the weighted Euclidean inner product of Formula (2) in Section 6.1, the desired inequality follows from the Cauchy-Schwarz inequality.
49. Using the inner product $\langle \mathbf{f}, \mathbf{g} \rangle = \int_0^1 f(x)g(x)dx$, part (a) follows from the Cauchy-Schwarz inequality and part (b) follows from the triangle inequality (part (a) of Theorem 6.2.2).
50. Squaring both sides of the Cauchy-Schwarz inequality and applying Definition 2 of Section 6.1 yields Formula (4).
51. (a) We are looking for all vectors $\mathbf{v} = (a, b)$ such that $\langle \mathbf{x}, \mathbf{v} \rangle = a + b$ is equal to $\langle T_A(\mathbf{x}), T_A(\mathbf{v}) \rangle = \begin{bmatrix} 2 \\ 0 \end{bmatrix} \cdot \begin{bmatrix} a+b \\ -a+b \end{bmatrix} = 2a + 2b$. The equation $a + b = 2a + 2b$ yields $a + b = 0$, i.e. $b = -a$. Vectors that satisfy $\langle \mathbf{x}, \mathbf{v} \rangle = \langle T_A(\mathbf{x}), T_A(\mathbf{v}) \rangle$ must have a form $a(1, -1)$ where a is an arbitrary scalar.
- (b) We are looking for all vectors $\mathbf{v} = (a, b)$ such that $\langle \mathbf{x}, \mathbf{v} \rangle = 2a + 3b$ is equal to $\langle T_A(\mathbf{x}), T_A(\mathbf{v}) \rangle = \begin{bmatrix} 2 \\ 0 \end{bmatrix} \cdot \begin{bmatrix} a+b \\ -a+b \end{bmatrix} = 4a + 4b$. The equation $2a + 3b = 4a + 4b$ yields $2a + b = 0$, i.e. $b = -2a$. Vectors that satisfy $\langle \mathbf{x}, \mathbf{v} \rangle = \langle T_A(\mathbf{x}), T_A(\mathbf{v}) \rangle$ must have a form $a(1, -2)$ where a is an arbitrary scalar.
52. (a) We are looking for all vectors $\mathbf{q} = a + bx + cx^2$ such that $\langle \mathbf{p}, \mathbf{q} \rangle = a + b$ is equal to $\langle T(\mathbf{p}), T(\mathbf{q}) \rangle = \langle 3, 3a - cx^2 \rangle = 9a$. The equation $a + b = 9a$ yields $b = 8a$. Vectors that satisfy $\langle \mathbf{p}, \mathbf{q} \rangle = \langle T(\mathbf{p}), T(\mathbf{q}) \rangle$ must have a form $a + 8ax + cx^2$ where a and c are arbitrary scalars.
- (b) We are looking for all vectors $\mathbf{q} = a + bx + cx^2$ such that $\langle \mathbf{p}, \mathbf{q} \rangle = p(-1)q(-1) + p(0)q(0) + p(1)q(1) = (0)(a - b + c) + (1)(a) + (2)(a + b + c) = 3a + 2b + 2c$ is equal to $\langle T(\mathbf{p}), T(\mathbf{q}) \rangle = \langle 3, 3a - cx^2 \rangle = (3)(3a - c) + (3)(3a) + (3)(3a - c) = 27a - 6c$. The equation $3a + 2b + 2c = 27a - 6c$ yields $b = 12a - 4c$. Vectors that satisfy $\langle \mathbf{p}, \mathbf{q} \rangle = \langle T(\mathbf{p}), T(\mathbf{q}) \rangle$ must have a form $a + (12a - 4c)x + cx^2$ where a and c are arbitrary scalars.

True-False Exercises

- (a) False. If $\mathbf{u}$ is orthogonal to every vector of a subspace W , then $\mathbf{u}$ is in $W^\perp$.
- (b) True. $W \cap W^\perp = \{\mathbf{0}\}$.

- (c) True. For any vector $\mathbf{w}$ in W , $\langle \mathbf{u} + \mathbf{v}, \mathbf{w} \rangle = \langle \mathbf{u}, \mathbf{w} \rangle + \langle \mathbf{v}, \mathbf{w} \rangle = 0$, so $\mathbf{u} + \mathbf{v}$ is in $W^\perp$.
- (d) True. For any vector $\mathbf{w}$ in W , $\langle k\mathbf{u}, \mathbf{w} \rangle = k\langle \mathbf{u}, \mathbf{w} \rangle = k(0) = 0$, so $k\mathbf{u}$ is in $W^\perp$.
- (e) False. If $\mathbf{u}$ and $\mathbf{v}$ are orthogonal $|\langle \mathbf{u}, \mathbf{v} \rangle| = |0| = 0$.
- (f) False. If $\mathbf{u}$ and $\mathbf{v}$ are orthogonal, $\|\mathbf{u} + \mathbf{v}\|^2 = \|\mathbf{u}\|^2 + \|\mathbf{v}\|^2$ thus $\mathbf{u} + \mathbf{v} = \sqrt{\|\mathbf{u}\|^2 + \|\mathbf{v}\|^2} \neq \|\mathbf{u}\| + \|\mathbf{v}\|$

6.3 Gram-Schmidt Process; QR-Decomposition

1. (a) $\langle (0, 1), (2, 0) \rangle = 0 + 0 = 0$;
 $\|(0, 1)\| = 1$; $\|(2, 0)\| = 2 \neq 1$;
 The set is orthogonal, but is not orthonormal.
- (b) $\left\langle \left(-\frac{1}{\sqrt{2}}, \frac{1}{\sqrt{2}}\right), \left(\frac{1}{\sqrt{2}}, \frac{1}{\sqrt{2}}\right) \right\rangle = -\frac{1}{2} + \frac{1}{2} = 0$;
 $\left\| \left(-\frac{1}{\sqrt{2}}, \frac{1}{\sqrt{2}}\right) \right\| = \sqrt{\frac{1}{2} + \frac{1}{2}} = 1$; $\left\| \left(\frac{1}{\sqrt{2}}, \frac{1}{\sqrt{2}}\right) \right\| = \sqrt{\frac{1}{2} + \frac{1}{2}} = 1$
 The set is orthogonal and orthonormal.
- (c) $\left\langle \left(-\frac{1}{\sqrt{2}}, -\frac{1}{\sqrt{2}}\right), \left(\frac{1}{\sqrt{2}}, \frac{1}{\sqrt{2}}\right) \right\rangle = -\frac{1}{2} - \frac{1}{2} = -1 \neq 0$;
 The set is not orthogonal (therefore, it is not orthonormal either).
- (d) $\langle (0, 0), (0, 1) \rangle = 0 + 0 = 0$;
 $\|(0, 0)\| = 0 \neq 1$; $\|(0, 1)\| = 1$;
 The set is orthogonal, but is not orthonormal.
2. (a) $\left\langle \left(\frac{1}{\sqrt{2}}, 0, \frac{1}{\sqrt{2}}\right), \left(\frac{1}{\sqrt{3}}, \frac{1}{\sqrt{3}}, -\frac{1}{\sqrt{3}}\right) \right\rangle = \frac{1}{\sqrt{6}} + 0 - \frac{1}{\sqrt{6}} = 0$; $\left\langle \left(\frac{1}{\sqrt{2}}, 0, \frac{1}{\sqrt{2}}\right), \left(-\frac{1}{\sqrt{2}}, 0, \frac{1}{\sqrt{2}}\right) \right\rangle = -\frac{1}{2} + 0 + \frac{1}{2} = 0$;
 $\left\langle \left(\frac{1}{\sqrt{3}}, \frac{1}{\sqrt{3}}, -\frac{1}{\sqrt{3}}\right), \left(-\frac{1}{\sqrt{2}}, 0, \frac{1}{\sqrt{2}}\right) \right\rangle = -\frac{1}{\sqrt{6}} - \frac{1}{\sqrt{6}} = -\frac{2}{\sqrt{6}} \neq 0$;
 The set is not orthogonal (therefore, it is not orthonormal either).
- (b) $\left\langle \left(\frac{2}{3}, -\frac{2}{3}, \frac{1}{3}\right), \left(\frac{2}{3}, \frac{1}{3}, -\frac{2}{3}\right) \right\rangle = \frac{4}{9} - \frac{2}{9} - \frac{2}{9} = 0$; $\left\langle \left(\frac{2}{3}, -\frac{2}{3}, \frac{1}{3}\right), \left(\frac{1}{3}, \frac{2}{3}, \frac{2}{3}\right) \right\rangle = \frac{2}{9} - \frac{4}{9} + \frac{2}{9} = 0$;
 $\left\langle \left(\frac{2}{3}, \frac{1}{3}, -\frac{2}{3}\right), \left(\frac{1}{3}, \frac{2}{3}, \frac{2}{3}\right) \right\rangle = \frac{2}{9} + \frac{2}{9} - \frac{4}{9} = 0$;
 $\left\| \left(\frac{2}{3}, -\frac{2}{3}, \frac{1}{3}\right) \right\| = \sqrt{\frac{4}{9} + \frac{4}{9} + \frac{1}{9}} = 1$; $\left\| \left(\frac{2}{3}, \frac{1}{3}, -\frac{2}{3}\right) \right\| = \sqrt{\frac{4}{9} + \frac{1}{9} + \frac{4}{9}} = 1$; $\left\| \left(\frac{1}{3}, \frac{2}{3}, \frac{2}{3}\right) \right\| = \sqrt{\frac{1}{9} + \frac{4}{9} + \frac{4}{9}} = 1$;
 The set is orthogonal and orthonormal.
- (c) $\left\langle (1, 0, 0), \left(0, \frac{1}{\sqrt{2}}, \frac{1}{\sqrt{2}}\right) \right\rangle = 0 + 0 + 0 = 0$; $\langle (1, 0, 0), (0, 0, 1) \rangle = 0 + 0 + 0 = 0$;
 $\left\langle \left(0, \frac{1}{\sqrt{2}}, \frac{1}{\sqrt{2}}\right), (0, 0, 1) \right\rangle = 0 + 0 + \frac{1}{\sqrt{2}} = \frac{1}{\sqrt{2}} \neq 0$;
 The set is not orthogonal (therefore, it is not orthonormal either).

$$(d) \quad \left\langle \left(\frac{1}{\sqrt{6}}, \frac{1}{\sqrt{6}}, -\frac{2}{\sqrt{6}} \right), \left(\frac{1}{\sqrt{2}}, -\frac{1}{\sqrt{2}}, 0 \right) \right\rangle = \frac{1}{\sqrt{12}} - \frac{1}{\sqrt{12}} + 0 = 0;$$

$$\left\| \left(\frac{1}{\sqrt{6}}, \frac{1}{\sqrt{6}}, -\frac{2}{\sqrt{6}} \right) \right\| = \sqrt{\frac{1}{6} + \frac{1}{6} + \frac{4}{6}} = 1; \quad \left\| \left(\frac{1}{\sqrt{2}}, -\frac{1}{\sqrt{2}}, 0 \right) \right\| = \sqrt{\frac{1}{2} + \frac{1}{2} + 0} = 1$$

The set is orthogonal and orthonormal.

$$3. \quad (a) \quad \langle p_1(x), p_2(x) \rangle = \frac{2}{3} \left(\frac{2}{3} \right) - \frac{2}{3} \left(\frac{1}{3} \right) + \frac{1}{3} \left(-\frac{2}{3} \right) = \frac{4}{9} - \frac{2}{9} - \frac{2}{9} = 0;$$

$$\langle p_1(x), p_3(x) \rangle = \frac{2}{3} \left(\frac{1}{3} \right) - \frac{2}{3} \left(\frac{2}{3} \right) + \frac{1}{3} \left(\frac{2}{3} \right) = \frac{2}{9} - \frac{4}{9} + \frac{2}{9} = 0;$$

$$\langle p_2(x), p_3(x) \rangle = \frac{2}{3} \left(\frac{1}{3} \right) + \frac{1}{3} \left(\frac{2}{3} \right) - \frac{2}{3} \left(\frac{2}{3} \right) = \frac{2}{9} + \frac{2}{9} - \frac{4}{9} = 0;$$

The set is orthogonal.

$$(b) \quad \langle p_1(x), p_2(x) \rangle = 1(0) + 0\left(\frac{1}{\sqrt{2}}\right) + 0\left(\frac{1}{\sqrt{2}}\right) = 0; \quad \langle p_1(x), p_3(x) \rangle = 1(0) + 0(0) + 0(1) = 0;$$

$$\langle p_2(x), p_3(x) \rangle = 0(0) + \frac{1}{\sqrt{2}}(0) + \frac{1}{\sqrt{2}}(1) = \frac{1}{\sqrt{2}} \neq 0;$$

The set is not orthogonal.

$$4. \quad (a) \quad \text{Denoting the matrices } A = \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}, B = \begin{bmatrix} 0 & \frac{2}{3} \\ \frac{1}{3} & -\frac{2}{3} \end{bmatrix}, C = \begin{bmatrix} 0 & \frac{2}{3} \\ -\frac{2}{3} & \frac{1}{3} \end{bmatrix}, \text{ and } D = \begin{bmatrix} 0 & \frac{1}{3} \\ \frac{2}{3} & \frac{2}{3} \end{bmatrix} \text{ we calculate}$$

$$\langle A, B \rangle = (1)(0) + (0)\left(\frac{2}{3}\right) + (0)\left(\frac{1}{3}\right) + (0)\left(-\frac{2}{3}\right) = 0;$$

$$\langle A, C \rangle = (1)(0) + (0)\left(\frac{2}{3}\right) + (0)\left(-\frac{2}{3}\right) + (0)\left(\frac{1}{3}\right) = 0;$$

$$\langle A, D \rangle = (1)(0) + (0)\left(\frac{1}{3}\right) + (0)\left(\frac{2}{3}\right) + (0)\left(\frac{2}{3}\right) = 0;$$

$$\langle B, C \rangle = (0)(0) + \left(\frac{2}{3}\right)\left(\frac{2}{3}\right) + \left(\frac{1}{3}\right)\left(-\frac{2}{3}\right) + \left(-\frac{2}{3}\right)\left(\frac{1}{3}\right) = 0;$$

$$\langle B, D \rangle = (0)(0) + \left(\frac{2}{3}\right)\left(\frac{1}{3}\right) + \left(\frac{1}{3}\right)\left(\frac{2}{3}\right) + \left(-\frac{2}{3}\right)\left(\frac{2}{3}\right) = 0;$$

$$\langle C, D \rangle = (0)(0) + \left(\frac{2}{3}\right)\left(\frac{1}{3}\right) + \left(-\frac{2}{3}\right)\left(\frac{2}{3}\right) + \left(\frac{1}{3}\right)\left(\frac{2}{3}\right) = 0;$$

The set is orthogonal.

$$(b) \quad \text{Denoting the matrices } A = \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}, B = \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix}, C = \begin{bmatrix} 0 & 0 \\ 1 & 1 \end{bmatrix}, \text{ and } D = \begin{bmatrix} 0 & 0 \\ 1 & -1 \end{bmatrix} \text{ we calculate}$$

$$\langle A, B \rangle = (1)(0) + (0)(1) + (0)(0) + (0)(0) = 0;$$

$$\langle A, C \rangle = (1)(0) + (0)(0) + (0)(1) + (0)(1) = 0;$$

$$\langle A, D \rangle = (1)(0) + (0)(0) + (0)(1) + (0)(-1) = 0;$$

$$\langle B, C \rangle = (0)(0) + (1)(0) + (0)(1) + (0)(1) = 0;$$

$$\langle B, D \rangle = (0)(0) + (1)(0) + (0)(1) + (0)(-1) = 0;$$

$$\langle C, D \rangle = (0)(0) + (0)(0) + (1)(1) + (1)(-1) = 0;$$

The set is orthogonal.

5. Let us denote the column vectors $\mathbf{u}_1 = (1, 0, -1)$, $\mathbf{u}_2 = (2, 0, 2)$, and $\mathbf{u}_3 = (0, 5, 0)$. These vectors are orthogonal since $\langle \mathbf{u}_1, \mathbf{u}_2 \rangle = 2 + 0 - 2 = 0$, $\langle \mathbf{u}_1, \mathbf{u}_3 \rangle = 0 + 0 + 0 = 0$, and $\langle \mathbf{u}_2, \mathbf{u}_3 \rangle = 0 + 0 + 0 = 0$.

It follows from Theorem 6.3.1 that the column vectors are linearly independent, therefore they form an

orthogonal basis for the column space of A . We proceed to normalize each column vector:

$$\frac{\mathbf{u}_1}{\|\mathbf{u}_1\|} = \frac{1}{\sqrt{1+0+1}}(1, 0, -1) = \left(\frac{1}{\sqrt{2}}, 0, -\frac{1}{\sqrt{2}}\right); \quad \frac{\mathbf{u}_2}{\|\mathbf{u}_2\|} = \frac{1}{\sqrt{4+0+4}}(2, 0, 2) = \left(\frac{2}{2\sqrt{2}}, 0, \frac{2}{2\sqrt{2}}\right) = \left(\frac{1}{\sqrt{2}}, 0, \frac{1}{\sqrt{2}}\right);$$

$$\frac{\mathbf{u}_3}{\|\mathbf{u}_3\|} = \frac{1}{\sqrt{0+25+0}}(0, 5, 0) = (0, 1, 0). \text{ A resulting orthonormal basis for the column space is}$$

$$\left\{\left(\frac{1}{\sqrt{2}}, 0, -\frac{1}{\sqrt{2}}\right), \left(\frac{1}{\sqrt{2}}, 0, \frac{1}{\sqrt{2}}\right), (0, 1, 0)\right\}.$$

6. Let us denote the column vectors $\mathbf{u}_1 = \left(\frac{1}{5}, \frac{1}{5}, \frac{1}{5}\right)$, $\mathbf{u}_2 = \left(-\frac{1}{2}, \frac{1}{2}, 0\right)$, and $\mathbf{u}_3 = \left(\frac{1}{3}, \frac{1}{3}, -\frac{2}{3}\right)$.

These vectors are orthogonal since $\langle \mathbf{u}_1, \mathbf{u}_2 \rangle = -\frac{1}{10} + \frac{1}{10} + 0 = 0$, $\langle \mathbf{u}_1, \mathbf{u}_3 \rangle = \frac{1}{15} + \frac{1}{15} - \frac{2}{15} = 0$, and

$\langle \mathbf{u}_2, \mathbf{u}_3 \rangle = -\frac{1}{6} + \frac{1}{6} + 0 = 0$. It follows from Theorem 6.3.1 that the column vectors are linearly independent, therefore they form an orthogonal basis for the column space of A . We proceed to normalize each column vector:

$$\frac{\mathbf{u}_1}{\|\mathbf{u}_1\|} = \frac{1}{\sqrt{\frac{1}{25} + \frac{1}{25} + \frac{1}{25}}} \left(\frac{1}{5}, \frac{1}{5}, \frac{1}{5}\right) = \frac{5}{\sqrt{3}} \left(\frac{1}{5}, \frac{1}{5}, \frac{1}{5}\right) = \left(\frac{1}{\sqrt{3}}, \frac{1}{\sqrt{3}}, \frac{1}{\sqrt{3}}\right);$$

$$\frac{\mathbf{u}_2}{\|\mathbf{u}_2\|} = \frac{1}{\sqrt{\frac{1}{4} + \frac{1}{4} + 0}} \left(-\frac{1}{2}, \frac{1}{2}, 0\right) = \sqrt{2} \left(-\frac{1}{2}, \frac{1}{2}, 0\right) = \left(-\frac{1}{\sqrt{2}}, \frac{1}{\sqrt{2}}, 0\right);$$

$$\frac{\mathbf{u}_3}{\|\mathbf{u}_3\|} = \frac{1}{\sqrt{\frac{1}{9} + \frac{1}{9} + \frac{4}{9}}} \left(\frac{1}{3}, \frac{1}{3}, -\frac{2}{3}\right) = \frac{3}{\sqrt{6}} \left(\frac{1}{3}, \frac{1}{3}, -\frac{2}{3}\right) = \left(\frac{1}{\sqrt{6}}, \frac{1}{\sqrt{6}}, -\frac{2}{\sqrt{6}}\right).$$

A resulting orthonormal basis for the column space is $\left\{\left(\frac{1}{\sqrt{3}}, \frac{1}{\sqrt{3}}, \frac{1}{\sqrt{3}}\right), \left(-\frac{1}{\sqrt{2}}, \frac{1}{\sqrt{2}}, 0\right), \left(\frac{1}{\sqrt{6}}, \frac{1}{\sqrt{6}}, -\frac{2}{\sqrt{6}}\right)\right\}$.

7. $\langle \mathbf{v}_1, \mathbf{v}_2 \rangle = -\frac{12}{25} + \frac{12}{25} + 0 = 0$; $\langle \mathbf{v}_1, \mathbf{v}_3 \rangle = 0 + 0 + 0 = 0$; $\langle \mathbf{v}_2, \mathbf{v}_3 \rangle = 0 + 0 + 0 = 0$;

$$\|\mathbf{v}_1\| = \sqrt{\frac{9}{25} + \frac{16}{25} + 0} = 1; \quad \|\mathbf{v}_2\| = \sqrt{\frac{16}{25} + \frac{9}{25} + 0} = 1; \quad \|\mathbf{v}_3\| = \sqrt{0 + 0 + 1} = 1;$$

Since this is an orthogonal set of nonzero vectors, it follows from Theorem 6.3.1 that the set is linearly independent. Because the number of vectors in the set matches $\dim(R^3) = 3$, this set forms a basis for R^3 by Theorem 4.6.4. This basis is orthonormal, so by Theorem 6.3.2(b),

$$\begin{aligned} \mathbf{u} &= \langle \mathbf{u}, \mathbf{v}_1 \rangle \mathbf{v}_1 + \langle \mathbf{u}, \mathbf{v}_2 \rangle \mathbf{v}_2 + \langle \mathbf{u}, \mathbf{v}_3 \rangle \mathbf{v}_3 \\ &= \left(-\frac{3}{5} - \frac{8}{5} + 0\right) \mathbf{v}_1 + \left(\frac{4}{5} - \frac{6}{5} + 0\right) \mathbf{v}_2 + (0 + 0 + 2) \mathbf{v}_3 \\ &= -\frac{11}{5} \mathbf{v}_1 - \frac{2}{5} \mathbf{v}_2 + 2 \mathbf{v}_3. \end{aligned}$$

8. It was shown in Exercise 7 that the vectors $\mathbf{v}_1$, $\mathbf{v}_2$, and $\mathbf{v}_3$ form an orthonormal basis for R^3 . By Theorem 6.3.2(b),

$$\begin{aligned} \mathbf{u} &= \langle \mathbf{u}, \mathbf{v}_1 \rangle \mathbf{v}_1 + \langle \mathbf{u}, \mathbf{v}_2 \rangle \mathbf{v}_2 + \langle \mathbf{u}, \mathbf{v}_3 \rangle \mathbf{v}_3 \\ &= \left(-\frac{9}{5} - \frac{28}{5} + 0\right) \mathbf{v}_1 + \left(\frac{12}{5} - \frac{21}{5} + 0\right) \mathbf{v}_2 + (0 + 0 + 4) \mathbf{v}_3 \\ &= -\frac{37}{5} \mathbf{v}_1 - \frac{9}{5} \mathbf{v}_2 + 4 \mathbf{v}_3. \end{aligned}$$

9. $\langle \mathbf{v}_1, \mathbf{v}_2 \rangle = 4 - 2 - 2 = 0$; $\langle \mathbf{v}_1, \mathbf{v}_3 \rangle = 2 - 4 + 2 = 0$; $\langle \mathbf{v}_2, \mathbf{v}_3 \rangle = 2 + 2 - 4 = 0$;

Since this is an orthogonal set of nonzero vectors, it follows from Theorem 6.3.1 that the set is linearly

independent. Because the number of vectors in the set matches $\dim(R^3) = 3$, this set forms a basis for R^3 by Theorem 4.6.4. By Theorem 6.3.2(a),

$$\begin{aligned}\mathbf{u} &= \frac{\langle \mathbf{u}, \mathbf{v}_1 \rangle}{\|\mathbf{v}_1\|^2} \mathbf{v}_1 + \frac{\langle \mathbf{u}, \mathbf{v}_2 \rangle}{\|\mathbf{v}_2\|^2} \mathbf{v}_2 + \frac{\langle \mathbf{u}, \mathbf{v}_3 \rangle}{\|\mathbf{v}_3\|^2} \mathbf{v}_3 \\ &= \frac{-2+0+2}{4+4+1} \mathbf{v}_1 + \frac{-2+0-4}{4+1+4} \mathbf{v}_2 + 0 \mathbf{v}_3 \\ &= 0 \mathbf{v}_1 - \frac{2}{3} \mathbf{v}_2 + \frac{1}{3} \mathbf{v}_3.\end{aligned}$$

$$\begin{aligned}10. \quad \langle \mathbf{v}_1, \mathbf{v}_2 \rangle &= -2 - 2 + 6 - 2 = 0 & \langle \mathbf{v}_1, \mathbf{v}_3 \rangle &= 1 - 2 + 0 + 1 = 0 & \langle \mathbf{v}_1, \mathbf{v}_4 \rangle &= 1 + 0 + 0 - 1 = 0 \\ \langle \mathbf{v}_2, \mathbf{v}_3 \rangle &= -2 + 4 + 0 - 2 = 0 & \langle \mathbf{v}_2, \mathbf{v}_4 \rangle &= -2 + 0 + 0 + 2 = 0 & \langle \mathbf{v}_3, \mathbf{v}_4 \rangle &= 1 + 0 + 0 - 1 = 0\end{aligned}$$

Since this is an orthogonal set of nonzero vectors, it follows from Theorem 6.3.1 that the set is linearly independent. Because the number of vectors in the set matches $\dim(R^4) = 4$, this set forms a basis for R^4 by Theorem 4.6.4. We use Theorem 6.3.2(a):

$$\begin{aligned}\mathbf{u} &= \frac{\langle \mathbf{u}, \mathbf{v}_1 \rangle}{\|\mathbf{v}_1\|^2} \mathbf{v}_1 + \frac{\langle \mathbf{u}, \mathbf{v}_2 \rangle}{\|\mathbf{v}_2\|^2} \mathbf{v}_2 + \frac{\langle \mathbf{u}, \mathbf{v}_3 \rangle}{\|\mathbf{v}_3\|^2} \mathbf{v}_3 + \frac{\langle \mathbf{u}, \mathbf{v}_4 \rangle}{\|\mathbf{v}_4\|^2} \mathbf{v}_4 \\ &= \frac{1-1+2-1}{1+1+4+1} \mathbf{v}_1 + \frac{-2+2+3+2}{4+4+9+4} \mathbf{v}_2 + \frac{1+2+0-1}{1+4+0+1} \mathbf{v}_3 + \frac{1+0+0+1}{1+0+0+1} \mathbf{v}_4 \\ &= \frac{1}{7} \mathbf{v}_1 + \frac{5}{21} \mathbf{v}_2 + \frac{1}{3} \mathbf{v}_3 + 1 \mathbf{v}_4.\end{aligned}$$

$$11. \quad (\mathbf{u})_S = \left(-\frac{11}{5}, -\frac{2}{5}, 2\right)$$

$$12. \quad (\mathbf{u})_S = \left(-\frac{37}{5}, -\frac{9}{5}, 4\right)$$

$$13. \quad (\mathbf{u})_S = \left(0, -\frac{2}{3}, \frac{1}{3}\right)$$

$$14. \quad (\mathbf{u})_S = \left(\frac{1}{7}, \frac{5}{21}, \frac{1}{3}, 1\right)$$

$$15. \quad (\mathbf{a}) \quad \|\mathbf{v}\| = \sqrt{\frac{9}{25} + \frac{16}{25}} = 1, \text{ so } \mathbf{v} \text{ forms an orthonormal basis for the line } W = \text{span}\{\mathbf{v}\}.$$

$$\mathbf{w}_1 = \text{proj}_W \mathbf{u} = \langle \mathbf{u}, \mathbf{v} \rangle \mathbf{v} = \left(-\frac{3}{5} + \frac{24}{5}\right) \left(\frac{3}{5}, \frac{4}{5}\right) = \frac{21}{5} \left(\frac{3}{5}, \frac{4}{5}\right) = \left(\frac{63}{25}, \frac{84}{25}\right)$$

$$(\mathbf{b}) \quad \mathbf{w}_2 = \mathbf{u} - \mathbf{w}_1 = (-1, 6) - \left(\frac{63}{25}, \frac{84}{25}\right) = \left(-\frac{88}{25}, \frac{66}{25}\right);$$

$$\mathbf{w}_2 \text{ is orthogonal to the line since } \langle \mathbf{w}_2, \mathbf{v} \rangle = -\frac{264}{125} + \frac{264}{125} = 0.$$

$$16. \quad (\mathbf{a}) \quad \|\mathbf{v}\| = \sqrt{\frac{25}{169} + \frac{144}{169}} = 1, \text{ so } \mathbf{v} \text{ forms an orthonormal basis for the line } W = \text{span}\{\mathbf{v}\}.$$

$$\mathbf{w}_1 = \text{proj}_W \mathbf{u} = \langle \mathbf{u}, \mathbf{v} \rangle \mathbf{v} = \left(\frac{10}{13} + \frac{36}{13}\right) \left(\frac{5}{13}, \frac{12}{13}\right) = \frac{46}{13} \left(\frac{5}{13}, \frac{12}{13}\right) = \left(\frac{230}{169}, \frac{552}{169}\right)$$

$$(\mathbf{b}) \quad \mathbf{w}_2 = \mathbf{u} - \mathbf{w}_1 = (2, 3) - \left(\frac{230}{169}, \frac{552}{169}\right) = \left(\frac{108}{169}, -\frac{45}{169}\right);$$

$$\mathbf{w}_2 \text{ is orthogonal to the line since } \langle \mathbf{w}_2, \mathbf{v} \rangle = \frac{540}{2197} - \frac{540}{2197} = 0.$$

17. (a) $\mathbf{w}_1 = \text{proj}_W \mathbf{u} = \frac{\langle \mathbf{u}, \mathbf{v} \rangle}{\|\mathbf{v}\|^2} \mathbf{v} = \frac{2+3}{1+1} (1, 1) = \frac{5}{2} (1, 1) = \left(\frac{5}{2}, \frac{5}{2}\right)$
- (b) $\mathbf{w}_2 = \mathbf{u} - \mathbf{w}_1 = (2, 3) - \left(\frac{5}{2}, \frac{5}{2}\right) = \left(-\frac{1}{2}, \frac{1}{2}\right)$;
 $\mathbf{w}_2$ is orthogonal to the line since $\langle \mathbf{w}_2, \mathbf{v} \rangle = -\frac{1}{2} + \frac{1}{2} = 0$.
18. (a) $\mathbf{w}_1 = \text{proj}_W \mathbf{u} = \frac{\langle \mathbf{u}, \mathbf{v} \rangle}{\|\mathbf{v}\|^2} \mathbf{v} = \frac{9-4}{9+16} (3, 4) = \frac{5}{25} (3, 4) = \left(\frac{3}{5}, \frac{4}{5}\right)$
- (b) $\mathbf{w}_2 = \mathbf{u} - \mathbf{w}_1 = (3, -1) - \left(\frac{3}{5}, \frac{4}{5}\right) = \left(\frac{12}{5}, -\frac{9}{5}\right)$;
 $\mathbf{w}_2$ is orthogonal to the line since $\langle \mathbf{w}_2, \mathbf{v} \rangle = \frac{36}{5} - \frac{36}{5} = 0$.
19. (a) $\langle \mathbf{v}_1, \mathbf{v}_2 \rangle = 0$ and $\|\mathbf{v}_1\| = \|\mathbf{v}_2\| = 1$, so $\{\mathbf{v}_1, \mathbf{v}_2\}$ is an orthonormal basis for the plane $W = \text{span}\{\mathbf{v}_1, \mathbf{v}_2\}$.
 $\mathbf{w}_1 = \text{proj}_W \mathbf{u} = \langle \mathbf{u}, \mathbf{v}_1 \rangle \mathbf{v}_1 + \langle \mathbf{u}, \mathbf{v}_2 \rangle \mathbf{v}_2 = 2\left(\frac{1}{3}, \frac{2}{3}, -\frac{2}{3}\right) + 4\left(\frac{2}{3}, \frac{1}{3}, \frac{2}{3}\right) = \left(\frac{10}{3}, \frac{8}{3}, \frac{4}{3}\right)$
- (b) $\mathbf{w}_2 = \mathbf{u} - \mathbf{w}_1 = (4, 2, 1) - \left(\frac{10}{3}, \frac{8}{3}, \frac{4}{3}\right) = \left(\frac{2}{3}, -\frac{2}{3}, -\frac{1}{3}\right)$;
 $\mathbf{w}_2$ is orthogonal to the plane since $\langle \mathbf{w}_2, \mathbf{v}_1 \rangle = \frac{2}{9} - \frac{4}{9} + \frac{2}{9} = 0$ and $\langle \mathbf{w}_2, \mathbf{v}_2 \rangle = \frac{4}{9} - \frac{2}{9} - \frac{2}{9} = 0$.
20. (a) $\langle \mathbf{v}_1, \mathbf{v}_2 \rangle = 0$ and $\|\mathbf{v}_1\| = \|\mathbf{v}_2\| = 1$, so $\{\mathbf{v}_1, \mathbf{v}_2\}$ is an orthonormal basis for the plane $W = \text{span}\{\mathbf{v}_1, \mathbf{v}_2\}$.
 $\mathbf{w}_1 = \text{proj}_W \mathbf{u} = \langle \mathbf{u}, \mathbf{v}_1 \rangle \mathbf{v}_1 + \langle \mathbf{u}, \mathbf{v}_2 \rangle \mathbf{v}_2 = -\frac{2}{\sqrt{6}} \left(\frac{1}{\sqrt{6}}, \frac{1}{\sqrt{6}}, -\frac{2}{\sqrt{6}}\right) + \frac{4}{\sqrt{3}} \left(\frac{1}{\sqrt{3}}, \frac{1}{\sqrt{3}}, \frac{1}{\sqrt{3}}\right)$
 $= \left(-\frac{2}{6}, -\frac{2}{6}, \frac{4}{6}\right) + \left(\frac{4}{3}, \frac{4}{3}, \frac{4}{3}\right) = \left(-\frac{1}{3}, -\frac{1}{3}, \frac{2}{3}\right) + \left(\frac{4}{3}, \frac{4}{3}, \frac{4}{3}\right) = (1, 1, 2)$
- (b) $\mathbf{w}_2 = \mathbf{u} - \mathbf{w}_1 = (3, -1, 2) - (1, 1, 2) = (2, -2, 0)$;
 $\mathbf{w}_2$ is orthogonal to the plane since $\langle \mathbf{w}_2, \mathbf{v}_1 \rangle = \frac{2}{\sqrt{6}} - \frac{2}{\sqrt{6}} + 0 = 0$ and $\langle \mathbf{w}_2, \mathbf{v}_2 \rangle = \frac{2}{\sqrt{3}} - \frac{2}{\sqrt{3}} = 0$.
21. (a) $\langle \mathbf{v}_1, \mathbf{v}_2 \rangle = 0$, so $\{\mathbf{v}_1, \mathbf{v}_2\}$ is an orthogonal basis for the plane $W = \text{span}\{\mathbf{v}_1, \mathbf{v}_2\}$.
 $\mathbf{w}_1 = \text{proj}_W \mathbf{u} = \frac{\langle \mathbf{u}, \mathbf{v}_1 \rangle}{\|\mathbf{v}_1\|^2} \mathbf{v}_1 + \frac{\langle \mathbf{u}, \mathbf{v}_2 \rangle}{\|\mathbf{v}_2\|^2} \mathbf{v}_2 = \frac{4}{6} (1, -2, 1) + \frac{2}{5} (2, 1, 0)$
 $= \left(\frac{2}{3}, -\frac{4}{3}, \frac{2}{3}\right) + \left(\frac{4}{5}, \frac{2}{5}, 0\right) = \left(\frac{22}{15}, -\frac{14}{15}, \frac{2}{3}\right)$
- (b) $\mathbf{w}_2 = \mathbf{u} - \mathbf{w}_1 = (1, 0, 3) - \left(\frac{22}{15}, -\frac{14}{15}, \frac{2}{3}\right) = \left(-\frac{7}{15}, \frac{14}{15}, \frac{7}{3}\right)$;
 $\mathbf{w}_2$ is orthogonal to the plane since $\langle \mathbf{w}_2, \mathbf{v}_1 \rangle = -\frac{7}{15} - \frac{28}{15} + \frac{7}{3} = \frac{-7-28+35}{15} = 0$ and
 $\langle \mathbf{w}_2, \mathbf{v}_2 \rangle = -\frac{14}{15} + \frac{14}{15} + 0 = 0$.
22. (a) $\langle \mathbf{v}_1, \mathbf{v}_2 \rangle = 0$, so $\{\mathbf{v}_1, \mathbf{v}_2\}$ is an orthogonal basis for the plane $W = \text{span}\{\mathbf{v}_1, \mathbf{v}_2\}$.
 $\mathbf{w}_1 = \text{proj}_W \mathbf{u} = \frac{\langle \mathbf{u}, \mathbf{v}_1 \rangle}{\|\mathbf{v}_1\|^2} \mathbf{v}_1 + \frac{\langle \mathbf{u}, \mathbf{v}_2 \rangle}{\|\mathbf{v}_2\|^2} \mathbf{v}_2 = \frac{7}{14} (3, 1, 2) + \frac{1}{3} (-1, 1, 1) = \left(\frac{3}{2}, \frac{1}{2}, 1\right) + \left(-\frac{1}{3}, \frac{1}{3}, \frac{1}{3}\right)$
 $= \left(\frac{7}{6}, \frac{5}{6}, \frac{4}{3}\right)$
- (b) $\mathbf{w}_2 = \mathbf{u} - \mathbf{w}_1 = (1, 0, 2) - \left(\frac{7}{6}, \frac{5}{6}, \frac{4}{3}\right) = \left(-\frac{1}{6}, -\frac{5}{6}, \frac{2}{3}\right)$;
 $\mathbf{w}_2$ is orthogonal to the plane since $\langle \mathbf{w}_2, \mathbf{v}_1 \rangle = -\frac{3}{6} - \frac{5}{6} + \frac{4}{3} = 0$ and $\langle \mathbf{w}_2, \mathbf{v}_2 \rangle = \frac{1}{6} - \frac{5}{6} + \frac{2}{3} = 0$.

$$\begin{aligned}
 23. \quad \text{proj}_W \mathbf{b} &= \frac{\langle \mathbf{b}, \mathbf{v}_1 \rangle}{\|\mathbf{v}_1\|^2} \mathbf{v}_1 + \frac{\langle \mathbf{b}, \mathbf{v}_2 \rangle}{\|\mathbf{v}_2\|^2} \mathbf{v}_2 = \frac{1+2+0-2}{1+1+1+1} (1, 1, 1, 1) + \frac{1+2+0+2}{1+1+1+1} (1, 1, -1, -1) \\
 &= \frac{1}{4} (1, 1, 1, 1) + \frac{5}{4} (1, 1, -1, -1) = \left(\frac{3}{2}, \frac{3}{2}, -1, -1 \right)
 \end{aligned}$$

$$\begin{aligned}
 24. \quad \text{proj}_W \mathbf{b} &= \frac{\langle \mathbf{b}, \mathbf{v}_1 \rangle}{\|\mathbf{v}_1\|^2} \mathbf{v}_1 + \frac{\langle \mathbf{b}, \mathbf{v}_2 \rangle}{\|\mathbf{v}_2\|^2} \mathbf{v}_2 = \frac{0+2+0+2}{0+1+16+1} (0, 1, -4, -1) + \frac{3+10+0-2}{9+25+1+1} (3, 5, 1, 1) \\
 &= \frac{2}{9} (0, 1, -4, -1) + \frac{11}{36} (3, 5, 1, 1) = \left(\frac{11}{12}, \frac{7}{4}, -\frac{7}{12}, \frac{1}{12} \right)
 \end{aligned}$$

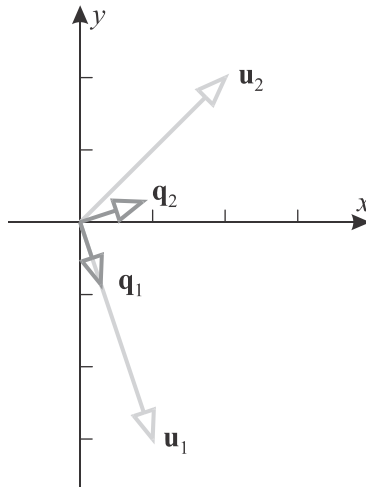
$$\begin{aligned}
 25. \quad \text{proj}_W \mathbf{b} &= \langle \mathbf{b}, \mathbf{v}_1 \rangle \mathbf{v}_1 + \langle \mathbf{b}, \mathbf{v}_2 \rangle \mathbf{v}_2 + \langle \mathbf{b}, \mathbf{v}_3 \rangle \mathbf{v}_3 \\
 &= \left(0 + \frac{2}{\sqrt{18}} + 0 + \frac{1}{\sqrt{18}} \right) \mathbf{v}_1 + \left(\frac{1}{2} + \frac{10}{6} + 0 - \frac{1}{6} \right) \mathbf{v}_2 + \left(\frac{1}{\sqrt{18}} + 0 + 0 + \frac{4}{\sqrt{18}} \right) \mathbf{v}_3 = \frac{3}{\sqrt{18}} \mathbf{v}_1 + 2 \mathbf{v}_2 + \frac{5}{\sqrt{18}} \mathbf{v}_3 \\
 &= \left(0, \frac{3}{18}, -\frac{12}{18}, -\frac{3}{18} \right) + \left(1, \frac{5}{3}, \frac{1}{3}, \frac{1}{3} \right) + \left(\frac{5}{18}, 0, \frac{5}{18}, -\frac{20}{18} \right) = \left(\frac{23}{18}, \frac{11}{6}, -\frac{1}{18}, -\frac{17}{18} \right)
 \end{aligned}$$

$$\begin{aligned}
 26. \quad \text{proj}_W \mathbf{b} &= \langle \mathbf{b}, \mathbf{v}_1 \rangle \mathbf{v}_1 + \langle \mathbf{b}, \mathbf{v}_2 \rangle \mathbf{v}_2 + \langle \mathbf{b}, \mathbf{v}_3 \rangle \mathbf{v}_3 \\
 &= \left(\frac{1}{2} + 1 + 0 - \frac{1}{2} \right) \mathbf{v}_1 + \left(\frac{1}{2} + 1 + 0 + \frac{1}{2} \right) \mathbf{v}_2 + \left(\frac{1}{2} - 1 + 0 + \frac{1}{2} \right) \mathbf{v}_3 = 1 \mathbf{v}_1 + 2 \mathbf{v}_2 + 0 \mathbf{v}_3 \\
 &= \left(\frac{1}{2}, \frac{1}{2}, \frac{1}{2}, \frac{1}{2} \right) + (1, 1, -1, -1) = \left(\frac{3}{2}, \frac{3}{2}, -\frac{1}{2}, -\frac{1}{2} \right)
 \end{aligned}$$

$$\begin{aligned}
 27. \quad \mathbf{v}_1 &= \mathbf{u}_1 = (1, -3) \\
 \mathbf{v}_2 &= \mathbf{u}_2 - \frac{\langle \mathbf{u}_2, \mathbf{v}_1 \rangle}{\|\mathbf{v}_1\|^2} \mathbf{v}_1 = (2, 2) - \frac{2-6}{10} (1, -3) = (2, 2) - \left(-\frac{2}{5}, \frac{6}{5} \right) = \left(\frac{12}{5}, \frac{4}{5} \right)
 \end{aligned}$$

An orthonormal basis is formed by the vectors $\mathbf{q}_1 = \frac{\mathbf{v}_1}{\|\mathbf{v}_1\|} = \frac{1}{\sqrt{10}} (1, -3) = \left(\frac{1}{\sqrt{10}}, -\frac{3}{\sqrt{10}} \right)$ and

$$\mathbf{q}_2 = \frac{\mathbf{v}_2}{\|\mathbf{v}_2\|} = \frac{1}{\sqrt{\frac{144}{25} + \frac{16}{25}}} \left(\frac{12}{5}, \frac{4}{5} \right) = \frac{1}{\sqrt{\frac{160}{25}}} \left(\frac{12}{5}, \frac{4}{5} \right) = \frac{5}{4\sqrt{10}} \left(\frac{12}{5}, \frac{4}{5} \right) = \left(\frac{3}{\sqrt{10}}, \frac{1}{\sqrt{10}} \right).$$

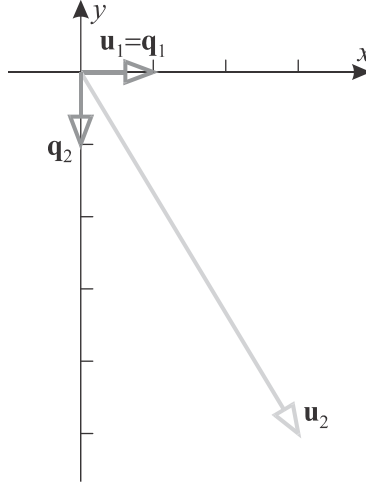


28. $\mathbf{v}_1 = \mathbf{u}_1 = (1, 0)$

$$\mathbf{v}_2 = \mathbf{u}_2 - \frac{\langle \mathbf{u}_2, \mathbf{v}_1 \rangle}{\|\mathbf{v}_1\|^2} \mathbf{v}_1 = (3, -5) - \frac{3+0}{1+1} (1, 0) = (3, -5) - (3, 0) = (0, -5)$$

An orthonormal basis is formed by the vectors $\mathbf{q}_1 = \frac{\mathbf{v}_1}{\|\mathbf{v}_1\|} = \frac{1}{\sqrt{1}} (1, 0) = (1, 0)$ and

$$\mathbf{q}_2 = \frac{\mathbf{v}_2}{\|\mathbf{v}_2\|} = \frac{1}{\sqrt{25}} (0, -5) = (0, -1).$$



29. $\mathbf{v}_1 = \mathbf{u}_1 = (1, 1, 1)$

$$\mathbf{v}_2 = \mathbf{u}_2 - \frac{\langle \mathbf{u}_2, \mathbf{v}_1 \rangle}{\|\mathbf{v}_1\|^2} \mathbf{v}_1 = (-1, 1, 0) - \frac{-1+1+0}{1+1+1} (1, 1, 1) = (-1, 1, 0) - 0(1, 1, 1) = (-1, 1, 0)$$

$$\begin{aligned} \mathbf{v}_3 &= \mathbf{u}_3 - \frac{\langle \mathbf{u}_3, \mathbf{v}_1 \rangle}{\|\mathbf{v}_1\|^2} \mathbf{v}_1 - \frac{\langle \mathbf{u}_3, \mathbf{v}_2 \rangle}{\|\mathbf{v}_2\|^2} \mathbf{v}_2 = (1, 2, 1) - \frac{1+2+1}{1+1+1} (1, 1, 1) - \frac{-1+2+0}{1+1+0} (-1, 1, 0) \\ &= (1, 2, 1) - \frac{4}{3} (1, 1, 1) - \frac{1}{2} (-1, 1, 0) = \left(\frac{1}{6}, \frac{1}{6}, -\frac{1}{3}\right) \end{aligned}$$

An orthonormal basis is formed by the vectors $\mathbf{q}_1 = \frac{\mathbf{v}_1}{\|\mathbf{v}_1\|} = \frac{1}{\sqrt{3}} (1, 1, 1) = \left(\frac{1}{\sqrt{3}}, \frac{1}{\sqrt{3}}, \frac{1}{\sqrt{3}}\right)$,

$$\mathbf{q}_2 = \frac{\mathbf{v}_2}{\|\mathbf{v}_2\|} = \frac{1}{\sqrt{2}} (-1, 1, 0) = \left(-\frac{1}{\sqrt{2}}, \frac{1}{\sqrt{2}}, 0\right), \text{ and } \mathbf{q}_3 = \frac{\mathbf{v}_3}{\|\mathbf{v}_3\|} = \frac{1}{1/\sqrt{6}} \left(\frac{1}{6}, \frac{1}{6}, -\frac{1}{3}\right) = \left(\frac{1}{\sqrt{6}}, \frac{1}{\sqrt{6}}, -\frac{2}{\sqrt{6}}\right).$$

30. $\mathbf{v}_1 = \mathbf{u}_1 = (1, 0, 0)$

$$\mathbf{v}_2 = \mathbf{u}_2 - \frac{\langle \mathbf{u}_2, \mathbf{v}_1 \rangle}{\|\mathbf{v}_1\|^2} \mathbf{v}_1 = (3, 7, -2) - \frac{3+0+0}{1+0+0} (1, 0, 0) = (3, 7, -2) - 3(1, 0, 0) = (0, 7, -2)$$

$$\begin{aligned} \mathbf{v}_3 &= \mathbf{u}_3 - \frac{\langle \mathbf{u}_3, \mathbf{v}_1 \rangle}{\|\mathbf{v}_1\|^2} \mathbf{v}_1 - \frac{\langle \mathbf{u}_3, \mathbf{v}_2 \rangle}{\|\mathbf{v}_2\|^2} \mathbf{v}_2 = (0, 4, 1) - \frac{0+0+0}{1+0+0} (1, 0, 0) - \frac{0+28-2}{0+49+4} (0, 7, -2) \\ &= (0, 4, 1) - 0(1, 0, 0) - \frac{26}{53} (0, 7, -2) = \left(0, \frac{30}{53}, \frac{105}{53}\right) \end{aligned}$$

An orthonormal basis is formed by the vectors $\mathbf{q}_1 = \frac{\mathbf{v}_1}{\|\mathbf{v}_1\|} = \frac{1}{1} (1, 0, 0) = (1, 0, 0)$,

$$\mathbf{q}_2 = \frac{\mathbf{v}_2}{\|\mathbf{v}_2\|} = \frac{1}{\sqrt{53}} (0, 7, -2) = \left(0, \frac{7}{\sqrt{53}}, -\frac{2}{\sqrt{53}}\right), \text{ and } \mathbf{q}_3 = \frac{\mathbf{v}_3}{\|\mathbf{v}_3\|} = \frac{1}{15/\sqrt{53}} \left(0, \frac{30}{53}, \frac{105}{53}\right) = \left(0, \frac{2}{\sqrt{53}}, \frac{7}{\sqrt{53}}\right).$$

31. First, transform the given basis into an orthogonal basis $\{\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3, \mathbf{v}_4\}$.

$$\mathbf{v}_1 = \mathbf{u}_1 = (0, 2, 1, 0)$$

$$\mathbf{v}_2 = \mathbf{u}_2 - \frac{\langle \mathbf{u}_2, \mathbf{v}_1 \rangle}{\|\mathbf{v}_1\|^2} \mathbf{v}_1 = (1, -1, 0, 0) - \frac{0-2+0+0}{5} (0, 2, 1, 0) = (1, -1, 0, 0) + \left(0, \frac{4}{5}, \frac{2}{5}, 0\right)$$

$$= (1, -\frac{1}{5}, \frac{2}{5}, 0)$$

$$\mathbf{v}_3 = \mathbf{u}_3 - \frac{\langle \mathbf{u}_3, \mathbf{v}_1 \rangle}{\|\mathbf{v}_1\|^2} \mathbf{v}_1 - \frac{\langle \mathbf{u}_3, \mathbf{v}_2 \rangle}{\|\mathbf{v}_2\|^2} \mathbf{v}_2 = (1, 2, 0, -1) - \frac{0+4+0+0}{5} (0, 2, 1, 0) - \frac{1-\frac{2}{5}+0+0}{\frac{6}{5}} (1, -\frac{1}{5}, \frac{2}{5}, 0)$$

$$= (1, 2, 0, -1) - (0, \frac{8}{5}, \frac{4}{5}, 0) - (\frac{1}{2}, -\frac{1}{10}, \frac{1}{5}, 0) = (\frac{1}{2}, \frac{1}{2}, -1, -1)$$

$$\mathbf{v}_4 = \mathbf{u}_4 - \frac{\langle \mathbf{u}_4, \mathbf{v}_1 \rangle}{\|\mathbf{v}_1\|^2} \mathbf{v}_1 - \frac{\langle \mathbf{u}_4, \mathbf{v}_2 \rangle}{\|\mathbf{v}_2\|^2} \mathbf{v}_2 - \frac{\langle \mathbf{u}_4, \mathbf{v}_3 \rangle}{\|\mathbf{v}_3\|^2} \mathbf{v}_3$$

$$= (1, 0, 0, 1) - \frac{0+0+0+0}{5} \mathbf{v}_1 - \frac{1+0+0+0}{\frac{6}{5}} (1, -\frac{1}{5}, \frac{2}{5}, 0) - \frac{\frac{1}{2}+0+0-1}{\frac{5}{2}} (\frac{1}{2}, \frac{1}{2}, -1, -1)$$

$$= (1, 0, 0, 1) - \frac{0+0+0+0}{5} \mathbf{v}_1 - \frac{1+0+0+0}{\frac{6}{5}} (1, -\frac{1}{5}, \frac{2}{5}, 0) - \frac{\frac{1}{2}+0+0-1}{\frac{5}{2}} (\frac{1}{2}, \frac{1}{2}, -1, -1) = (\frac{4}{15}, \frac{4}{15}, -\frac{8}{15}, \frac{4}{5})$$

An orthonormal basis is formed by the vectors $\mathbf{q}_1 = \frac{\mathbf{v}_1}{\|\mathbf{v}_1\|} = \frac{(0, 2, 1, 0)}{\sqrt{5}} = (0, \frac{2}{\sqrt{5}}, \frac{1}{\sqrt{5}}, 0)$,

$$\mathbf{q}_2 = \frac{\mathbf{v}_2}{\|\mathbf{v}_2\|} = \frac{(1, -\frac{1}{5}, \frac{2}{5}, 0)}{\frac{\sqrt{30}}{5}} = (\frac{5}{\sqrt{30}}, -\frac{1}{\sqrt{30}}, \frac{2}{\sqrt{30}}, 0), \quad \mathbf{q}_3 = \frac{\mathbf{v}_3}{\|\mathbf{v}_3\|} = \frac{(\frac{1}{2}, \frac{1}{2}, -1, -1)}{\frac{\sqrt{10}}{2}} = (\frac{1}{\sqrt{10}}, \frac{1}{\sqrt{10}}, -\frac{2}{\sqrt{10}}, -\frac{2}{\sqrt{10}}),$$

$$\mathbf{q}_4 = \frac{\mathbf{v}_4}{\|\mathbf{v}_4\|} = \frac{(\frac{4}{15}, \frac{4}{15}, -\frac{8}{15}, \frac{4}{5})}{\frac{4}{\sqrt{15}}} = (\frac{1}{\sqrt{15}}, \frac{1}{\sqrt{15}}, -\frac{2}{\sqrt{15}}, \frac{3}{\sqrt{15}}).$$

32. We begin by forming a matrix whose columns are the three given vectors: $\begin{bmatrix} 0 & -1 & -1 \\ 1 & 0 & 1 \\ 2 & 1 & 3 \end{bmatrix}$. The reduced row

echelon form of this matrix is $\begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 1 \\ 0 & 0 & 0 \end{bmatrix}$. By Theorem 4.8.5(b), vectors $\mathbf{u}_1 = (0, 1, 2)$ and

$\mathbf{u}_2 = (-1, 0, 1)$ form a basis for the span of the three given vectors (the third vector can be expressed as a linear combination of the first two).

We proceed to use the Gram-Schmidt process to transform the basis $\{\mathbf{u}_1, \mathbf{u}_2\}$ to an orthonormal basis.

$$\mathbf{v}_1 = \mathbf{u}_1 = (0, 1, 2)$$

$$\mathbf{v}_2 = \mathbf{u}_2 - \frac{\langle \mathbf{u}_2, \mathbf{v}_1 \rangle}{\|\mathbf{v}_1\|^2} \mathbf{v}_1 = (-1, 0, 1) - \frac{0+0+2}{0+1+4} (0, 1, 2) = (-1, 0, 1) - \frac{2}{5} (0, 1, 2) = (-1, -\frac{2}{5}, \frac{1}{5})$$

An orthonormal basis is formed by the vectors

$$\mathbf{q}_1 = \frac{\mathbf{v}_1}{\|\mathbf{v}_1\|} = \frac{1}{\sqrt{5}} (0, 1, 2) = (0, \frac{1}{\sqrt{5}}, \frac{2}{\sqrt{5}}) \text{ and } \mathbf{q}_2 = \frac{\mathbf{v}_2}{\|\mathbf{v}_2\|} = \frac{1}{\sqrt{6/5}} (-1, -\frac{2}{5}, \frac{1}{5}) = (-\frac{5}{\sqrt{30}}, -\frac{2}{\sqrt{30}}, \frac{1}{\sqrt{30}}).$$

33. From Exercise 23, $\mathbf{w}_1 = \text{proj}_W \mathbf{b} = (\frac{3}{2}, \frac{3}{2}, -1, -1)$, so $\mathbf{w}_2 = \mathbf{b} - \text{proj}_W \mathbf{b} = (-\frac{1}{2}, \frac{1}{2}, 1, -1)$.

34. From Exercise 25, $\mathbf{w}_1 = \text{proj}_W \mathbf{b} = (\frac{23}{18}, \frac{11}{6}, -\frac{1}{18}, -\frac{17}{18})$, so $\mathbf{w}_2 = \mathbf{b} - \mathbf{w}_1 = (-\frac{5}{18}, \frac{1}{6}, \frac{1}{18}, -\frac{1}{18})$.

35. Let W be the plane spanned by the vectors $\mathbf{u}_1$ and $\mathbf{u}_2$.

$$\mathbf{v}_1 = \mathbf{u}_1 = (1, 1, 1)$$

$$\mathbf{v}_2 = \mathbf{u}_2 - \frac{\langle \mathbf{u}_2, \mathbf{v}_1 \rangle}{\|\mathbf{v}_1\|^2} \mathbf{v}_1 = (2, 0, -1) - \frac{2+0-1}{1+1+1} (1, 1, 1) = (2, 0, -1) - (\frac{1}{3}, \frac{1}{3}, \frac{1}{3}) = (\frac{5}{3}, -\frac{1}{3}, -\frac{4}{3})$$

$$\mathbf{w}_1 = \text{proj}_W \mathbf{w} = \frac{\langle \mathbf{w}, \mathbf{v}_1 \rangle}{\|\mathbf{v}_1\|^2} \mathbf{v}_1 + \frac{\langle \mathbf{w}, \mathbf{v}_2 \rangle}{\|\mathbf{v}_2\|^2} \mathbf{v}_2 = \frac{1+2+3}{3} \mathbf{v}_1 + \frac{\frac{5}{3}-\frac{2}{3}-\frac{12}{3}}{\frac{42}{9}} \mathbf{v}_2 = 2(1, 1, 1) - \frac{9}{14} (\frac{5}{3}, -\frac{1}{3}, -\frac{4}{3})$$

$$= (2, 2, 2) - \left(\frac{15}{14}, -\frac{3}{14}, -\frac{6}{7}\right) = \left(\frac{13}{14}, \frac{31}{14}, \frac{20}{7}\right)$$

$$\mathbf{w}_2 = \mathbf{w} - \mathbf{w}_1 = (1, 2, 3) - \left(\frac{13}{14}, \frac{31}{14}, \frac{20}{7}\right) = \left(\frac{1}{14}, -\frac{3}{14}, \frac{1}{7}\right)$$

36. The vectors $\mathbf{u}_1$ and $\mathbf{u}_2$ are not orthogonal, therefore we begin by using the Gram-Schmidt process to transform the basis $\{\mathbf{u}_1, \mathbf{u}_2\}$ to an orthonormal basis.

$$\mathbf{v}_1 = \mathbf{u}_1 = (-1, 0, 1, 2)$$

$$\mathbf{v}_2 = \mathbf{u}_2 - \frac{\langle \mathbf{u}_2, \mathbf{v}_1 \rangle}{\|\mathbf{v}_1\|^2} \mathbf{v}_1 = (0, 1, 0, 1) - \frac{0+0+0+2}{1+0+1+4} (-1, 0, 1, 2) = (0, 1, 0, 1) - \frac{1}{3} (-1, 0, 1, 2) = \left(\frac{1}{3}, 1, -\frac{1}{3}, \frac{1}{3}\right)$$

An orthonormal basis is formed by the vectors

$$\mathbf{q}_1 = \frac{\mathbf{v}_1}{\|\mathbf{v}_1\|} = \frac{1}{\sqrt{6}} (-1, 0, 1, 2) = \left(-\frac{1}{\sqrt{6}}, 0, \frac{1}{\sqrt{6}}, \frac{2}{\sqrt{6}}\right) \text{ and } \mathbf{q}_2 = \frac{\mathbf{v}_2}{\|\mathbf{v}_2\|} = \frac{1}{\sqrt{4/3}} \left(\frac{1}{3}, 1, -\frac{1}{3}, \frac{1}{3}\right) = \left(\frac{1}{2\sqrt{3}}, \frac{3}{2\sqrt{3}}, -\frac{1}{2\sqrt{3}}, \frac{1}{2\sqrt{3}}\right).$$

$$\mathbf{w}_1 = \text{proj}_{\mathbf{w}} \mathbf{w} = \langle \mathbf{w}, \mathbf{q}_1 \rangle \mathbf{q}_1 + \langle \mathbf{w}, \mathbf{q}_2 \rangle \mathbf{q}_2 = \frac{1+0+6+0}{\sqrt{6}} \left(-\frac{1}{\sqrt{6}}, 0, \frac{1}{\sqrt{6}}, \frac{2}{\sqrt{6}}\right) + \frac{-1+6-6+0}{2\sqrt{3}} \left(\frac{1}{2\sqrt{3}}, \frac{3}{2\sqrt{3}}, -\frac{1}{2\sqrt{3}}, \frac{1}{2\sqrt{3}}\right)$$

$$= \left(-\frac{7}{6}, 0, \frac{7}{6}, \frac{7}{3}\right) + \left(-\frac{1}{12}, -\frac{1}{4}, \frac{1}{12}, -\frac{1}{12}\right) = \left(-\frac{5}{4}, -\frac{1}{4}, \frac{5}{4}, \frac{9}{4}\right)$$

$$\mathbf{w}_2 = \mathbf{w} - \mathbf{w}_1 = (-1, 2, 6, 0) - \left(-\frac{5}{4}, -\frac{1}{4}, \frac{5}{4}, \frac{9}{4}\right) = \left(\frac{1}{4}, \frac{9}{4}, \frac{19}{4}, -\frac{9}{4}\right).$$

37. First, transform the given basis into an orthogonal basis $\{\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3\}$.

$$\mathbf{v}_1 = \mathbf{u}_1 = (1, 1, 1)$$

$$\|\mathbf{v}_1\| = \sqrt{\langle \mathbf{v}_1, \mathbf{v}_1 \rangle} = \sqrt{1^2 + 2(1)^2 + 3(1)^2} = \sqrt{1+2+3} = \sqrt{6}$$

$$\mathbf{v}_2 = \mathbf{u}_2 - \frac{\langle \mathbf{u}_2, \mathbf{v}_1 \rangle}{\|\mathbf{v}_1\|^2} \mathbf{v}_1 = (1, 1, 0) - \frac{1(1)+2(1)(1)+3(0)(1)}{6} (1, 1, 1) = (1, 1, 0) - \frac{1}{2} (1, 1, 1) = \left(\frac{1}{2}, \frac{1}{2}, -\frac{1}{2}\right)$$

$$\|\mathbf{v}_2\| = \sqrt{\langle \mathbf{v}_2, \mathbf{v}_2 \rangle} = \sqrt{\left(\frac{1}{2}\right)^2 + 2\left(\frac{1}{2}\right)^2 + 3\left(-\frac{1}{2}\right)^2} = \sqrt{6\left(\frac{1}{4}\right)} = \frac{\sqrt{6}}{2}$$

$$\mathbf{v}_3 = \mathbf{u}_3 - \frac{\langle \mathbf{u}_3, \mathbf{v}_1 \rangle}{\|\mathbf{v}_1\|^2} \mathbf{v}_1 - \frac{\langle \mathbf{u}_3, \mathbf{v}_2 \rangle}{\|\mathbf{v}_2\|^2} \mathbf{v}_2 = (1, 0, 0) - \frac{1+0+0}{6} (1, 1, 1) - \frac{\frac{1}{2}+0+0}{\frac{6}{4}} \left(\frac{1}{2}, \frac{1}{2}, -\frac{1}{2}\right)$$

$$= (1, 0, 0) - \left(\frac{1}{6}, \frac{1}{6}, \frac{1}{6}\right) - \left(\frac{1}{6}, \frac{1}{6}, -\frac{1}{6}\right) = \left(\frac{2}{3}, -\frac{1}{3}, 0\right)$$

$$\|\mathbf{v}_3\| = \sqrt{\langle \mathbf{v}_3, \mathbf{v}_3 \rangle} = \sqrt{\left(\frac{2}{3}\right)^2 + 2\left(-\frac{1}{3}\right)^2 + 3(0)^2} = \sqrt{\frac{4}{9} + \frac{2}{9}} = \frac{\sqrt{6}}{3}$$

The orthonormal basis is $\mathbf{q}_1 = \frac{\mathbf{v}_1}{\|\mathbf{v}_1\|} = \frac{(1, 1, 1)}{\sqrt{6}} = \left(\frac{1}{\sqrt{6}}, \frac{1}{\sqrt{6}}, \frac{1}{\sqrt{6}}\right)$, $\mathbf{q}_2 = \frac{\mathbf{v}_2}{\|\mathbf{v}_2\|} = \frac{(\frac{1}{2}, \frac{1}{2}, -\frac{1}{2})}{\frac{\sqrt{6}}{2}} = \left(\frac{1}{\sqrt{6}}, \frac{1}{\sqrt{6}}, -\frac{1}{\sqrt{6}}\right)$, and

$$\mathbf{q}_3 = \frac{\mathbf{v}_3}{\|\mathbf{v}_3\|} = \frac{(\frac{2}{3}, -\frac{1}{3}, 0)}{\frac{\sqrt{6}}{3}} = \left(\frac{2}{\sqrt{6}}, -\frac{1}{\sqrt{6}}, 0\right).$$

38. Let us denote $\mathbf{w}_1 = (1, 0)$ and $\mathbf{w}_2 = (0, 1)$. The set is orthogonal because $\langle \mathbf{w}_1, \mathbf{w}_2 \rangle = 4(1)(0) + (0)(1) = 0$.

$$\frac{\mathbf{w}_1}{\|\mathbf{w}_1\|} = \frac{\mathbf{w}_1}{\sqrt{\langle \mathbf{w}_1, \mathbf{w}_1 \rangle}} = \frac{1}{\sqrt{4(1)(1)+(0)(0)}} (1, 0) = \left(\frac{1}{2}, 0\right); \quad \frac{\mathbf{w}_2}{\|\mathbf{w}_2\|} = \frac{\mathbf{w}_2}{\sqrt{\langle \mathbf{w}_2, \mathbf{w}_2 \rangle}} = \frac{1}{\sqrt{4(0)(0)+(1)(1)}} (0, 1) = (0, 1)$$

By normalizing the given orthogonal set, we obtained an orthonormal set $\left\{\left(\frac{1}{2}, 0\right), (0, 1)\right\}$.

39. For example, $\mathbf{x} = \left(\frac{1}{\sqrt{3}}, 0\right)$ and $\mathbf{y} = \left(0, \frac{1}{\sqrt{2}}\right)$.

40. The line through the origin making $\frac{\pi}{6}$ angle with the x -axis is a subspace of R^2 spanned by the vector

$\mathbf{u} = (\cos \frac{\pi}{6}, \sin \frac{\pi}{6}) = (\frac{\sqrt{3}}{2}, \frac{1}{2})$. Since this vector has norm 1, the orthogonal projection of $\mathbf{x} = (1, 5)$ onto

$W = \text{span}\{\mathbf{u}\}$ is $\text{proj}_W \mathbf{x} = \langle \mathbf{x}, \mathbf{u} \rangle \mathbf{u} = (\frac{\sqrt{3}}{2} + \frac{5}{2})(\frac{\sqrt{3}}{2}, \frac{1}{2}) = (\frac{3+5\sqrt{3}}{4}, \frac{5+\sqrt{3}}{4})$.

41. (a) By inspection, $\mathbf{v}'_1 = \mathbf{v}_1 + \mathbf{v}_2$ and $\mathbf{v}'_2 = \mathbf{v}_1 - 2\mathbf{v}_2$ so $\mathbf{v}'_1$ and $\mathbf{v}'_2$ are in W . The dimension of W is 2 since $\{\mathbf{v}_1, \mathbf{v}_2\}$ is an orthogonal basis for W . By Theorem 6.3.1, $\{\mathbf{v}'_1, \mathbf{v}'_2\}$ is linearly independent, so by Theorem 4.6.4 it is a basis for W , hence it spans W .

(b) Calculating $\text{proj}_W \mathbf{u}$ using $\{\mathbf{v}_1, \mathbf{v}_2\}$ we obtain

$$\frac{\langle \mathbf{u}, \mathbf{v}_1 \rangle}{\|\mathbf{v}_1\|^2} \mathbf{v}_1 + \frac{\langle \mathbf{u}, \mathbf{v}_2 \rangle}{\|\mathbf{v}_2\|^2} \mathbf{v}_2 = \frac{-3+0+7}{1+0+1} (1, 0, 1) + \frac{0+1+0}{0+1+0} (0, 1, 0) = (2, 0, 2) + (0, 1, 0) = (2, 1, 2).$$

Calculating $\text{proj}_W \mathbf{u}$ using $\{\mathbf{v}'_1, \mathbf{v}'_2\}$ instead yields the same vector:

$$\frac{\langle \mathbf{u}, \mathbf{v}'_1 \rangle}{\|\mathbf{v}'_1\|^2} \mathbf{v}'_1 + \frac{\langle \mathbf{u}, \mathbf{v}'_2 \rangle}{\|\mathbf{v}'_2\|^2} \mathbf{v}'_2 = \frac{-3+1+7}{1+1+1} (1, 1, 1) + \frac{-3-2+7}{1+4+1} (1, -2, 1) = (\frac{5}{3}, \frac{5}{3}, \frac{5}{3}) + (\frac{1}{3}, -\frac{2}{3}, \frac{1}{3}) = (2, 1, 2).$$

42. The first three Legendre polynomials are $q_1(x) = 1$, $q_2(x) = x$, and $q_3(x) = \frac{1}{2}(3x^2 - 1) = -\frac{1}{2} + \frac{3}{2}x^2$.

By Example 8 and the Remark that follows it, these polynomials form an orthogonal set in the vector space

P_2 with the inner product $\langle \mathbf{p}, \mathbf{q} \rangle = \int_{-1}^1 p(x)q(x)dx$.

We have $\|\mathbf{q}_1\|^2 = \int_{-1}^1 (1)^2 dx = x \Big|_{-1}^1 = 2$, $\|\mathbf{q}_2\|^2 = \int_{-1}^1 (x)^2 dx = \frac{x^3}{3} \Big|_{-1}^1 = \frac{2}{3}$, and

$$\|\mathbf{q}_3\|^2 = \int_{-1}^1 \left(-\frac{1}{2} + \frac{3}{2}x^2\right)^2 dx = \left(\frac{x}{4} - \frac{x^3}{2} + \frac{9x^5}{20}\right) \Big|_{-1}^1 = \frac{2}{5}.$$

(a) For $p(x) = 1 + x + 4x^2$, Theorem 6.3.2(a) yields

$$\begin{aligned} \mathbf{p} &= \frac{\langle \mathbf{p}, \mathbf{q}_1 \rangle}{\|\mathbf{q}_1\|^2} \mathbf{q}_1 + \frac{\langle \mathbf{p}, \mathbf{q}_2 \rangle}{\|\mathbf{q}_2\|^2} \mathbf{q}_2 + \frac{\langle \mathbf{p}, \mathbf{q}_3 \rangle}{\|\mathbf{q}_3\|^2} \mathbf{q}_3 \\ &= \frac{\int_{-1}^1 (1+x+4x^2)dx}{2} \mathbf{q}_1 + \frac{\int_{-1}^1 (x+x^2+4x^3)dx}{\frac{2}{3}} \mathbf{q}_2 + \frac{\int_{-1}^1 \left(-\frac{1}{2} - \frac{x}{2} - \frac{x^2}{2} + \frac{3x^3}{2} + 6x^4\right)dx}{\frac{2}{5}} \mathbf{q}_3 \\ &= \frac{1}{2} \left(x + \frac{x^2}{2} + \frac{4x^3}{3} \right) \Big|_{-1}^1 \mathbf{q}_1 + \frac{3}{2} \left(\frac{x^2}{2} + \frac{x^3}{3} + x^4 \right) \Big|_{-1}^1 \mathbf{q}_2 + \frac{5}{2} \left(-\frac{x}{2} - \frac{x^2}{4} - \frac{x^3}{6} + \frac{3x^4}{8} + \frac{6x^5}{5} \right) \Big|_{-1}^1 \mathbf{q}_3 \\ &= \left(\frac{1}{2} \right) \left(\frac{14}{3} \right) \mathbf{q}_1 + \left(\frac{3}{2} \right) \left(\frac{2}{3} \right) \mathbf{q}_2 + \left(\frac{5}{2} \right) \left(\frac{16}{15} \right) \mathbf{q}_3 \\ &= \frac{7}{3} \mathbf{q}_1 + 1 \mathbf{q}_2 + \frac{8}{3} \mathbf{q}_3 \end{aligned}$$

(b) For $p(x) = 2 - 7x^2$, Theorem 6.3.2(a) yields

$$\mathbf{p} = \frac{\langle \mathbf{p}, \mathbf{q}_1 \rangle}{\|\mathbf{q}_1\|^2} \mathbf{q}_1 + \frac{\langle \mathbf{p}, \mathbf{q}_2 \rangle}{\|\mathbf{q}_2\|^2} \mathbf{q}_2 + \frac{\langle \mathbf{p}, \mathbf{q}_3 \rangle}{\|\mathbf{q}_3\|^2} \mathbf{q}_3$$

$$\begin{aligned}
&= \frac{\int_{-1}^1 (2-7x^2) dx}{2} \mathbf{q}_1 + \frac{\int_{-1}^1 (2x-7x^3) dx}{\frac{2}{3}} \mathbf{q}_2 + \frac{\int_{-1}^1 \left(-1 + \frac{13x^2}{2} - \frac{21x^4}{2}\right) dx}{\frac{2}{5}} \mathbf{q}_3 \\
&= \frac{1}{2} \left(2x - \frac{7x^3}{3}\right) \Big|_{-1}^1 \mathbf{q}_1 + \frac{3}{2} \left(x^2 - \frac{7x^4}{4}\right) \Big|_{-1}^1 \mathbf{q}_2 + \frac{5}{2} \left(-x + \frac{13x^3}{6} - \frac{21x^5}{10}\right) \Big|_{-1}^1 \mathbf{q}_3 \\
&= \left(\frac{1}{2}\right) \left(-\frac{2}{3}\right) \mathbf{q}_1 + \left(\frac{3}{2}\right) (0) \mathbf{q}_2 + \left(\frac{5}{2}\right) \left(-\frac{28}{15}\right) \mathbf{q}_3 \\
&= -\frac{1}{3} \mathbf{q}_1 + 0 \mathbf{q}_2 - \frac{14}{3} \mathbf{q}_3
\end{aligned}$$

(c) For $p(x) = 4 + 3x$, Theorem 6.3.2(a) yields

$$\begin{aligned}
\mathbf{p} &= \frac{\langle \mathbf{p}, \mathbf{q}_1 \rangle}{\|\mathbf{q}_1\|^2} \mathbf{q}_1 + \frac{\langle \mathbf{p}, \mathbf{q}_2 \rangle}{\|\mathbf{q}_2\|^2} \mathbf{q}_2 + \frac{\langle \mathbf{p}, \mathbf{q}_3 \rangle}{\|\mathbf{q}_3\|^2} \mathbf{q}_3 \\
&= \frac{\int_{-1}^1 (4+3x) dx}{2} \mathbf{q}_1 + \frac{\int_{-1}^1 (4x+3x^2) dx}{\frac{2}{3}} \mathbf{q}_2 + \frac{\int_{-1}^1 \left(-2 - \frac{3x}{2} + 6x^2 + \frac{9x^3}{2}\right) dx}{\frac{2}{5}} \mathbf{q}_3 \\
&= \left(\frac{1}{2}\right) \left(4x + \frac{3x^2}{2}\right) \Big|_{-1}^1 \mathbf{q}_1 + \frac{3}{2} \left(2x^2 + x^3\right) \Big|_{-1}^1 \mathbf{q}_2 + \frac{5}{2} \left(-2x - \frac{3x^2}{4} + 2x^3 + \frac{9x^4}{8}\right) \Big|_{-1}^1 \mathbf{q}_3 \\
&= \left(\frac{1}{2}\right) (8) \mathbf{q}_1 + \left(\frac{3}{2}\right) (2) \mathbf{q}_2 + \left(\frac{5}{2}\right) (0) \mathbf{q}_3 \\
&= 4\mathbf{q}_1 + 3\mathbf{q}_2 + 0\mathbf{q}_3
\end{aligned}$$

43. First transform the basis $S = \{\mathbf{p}_1, \mathbf{p}_2, \mathbf{p}_3\} = \{1, x, x^2\}$ into an orthogonal basis $\{\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3\}$.

$$\mathbf{v}_1 = \mathbf{p}_1 = 1$$

$$\|\mathbf{v}_1\| = \sqrt{\langle \mathbf{v}_1, \mathbf{v}_1 \rangle} = \sqrt{\int_0^1 1^2 dx} = \sqrt{x} \Big|_0^1 = 1$$

$$\langle \mathbf{p}_2, \mathbf{v}_1 \rangle = \int_0^1 x \cdot 1 dx = \frac{1}{2} x^2 \Big|_0^1 = \frac{1}{2}$$

$$\mathbf{v}_2 = \mathbf{p}_2 - \frac{\langle \mathbf{p}_2, \mathbf{v}_1 \rangle}{\|\mathbf{v}_1\|^2} \mathbf{v}_1 = x - \frac{\frac{1}{2}}{1} (1) = x - \frac{1}{2} (1) = -\frac{1}{2} + x$$

$$\|\mathbf{v}_2\|^2 = \langle \mathbf{v}_2, \mathbf{v}_2 \rangle = \int_0^1 \left(-\frac{1}{2} + x\right)^2 dx = \int_0^1 \left(\frac{1}{4} - x + x^2\right) dx = \left(\frac{1}{4}x - \frac{1}{2}x^2 + \frac{1}{3}x^3\right) \Big|_0^1 = \frac{1}{12}$$

$$\|\mathbf{v}_2\| = \sqrt{\frac{1}{12}} = \frac{1}{2\sqrt{3}}$$

$$\langle \mathbf{p}_3, \mathbf{v}_1 \rangle = \int_0^1 x^2 \cdot 1 dx = \frac{1}{3} x^3 \Big|_0^1 = \frac{1}{3}$$

$$\langle \mathbf{p}_3, \mathbf{v}_2 \rangle = \int_0^1 x^2 \left(-\frac{1}{2} + x\right) dx = \int_0^1 \left(-\frac{1}{2}x^2 + x^3\right) dx = \left(-\frac{1}{6}x^3 + \frac{1}{4}x^4\right) \Big|_0^1 = \frac{1}{12}$$

$$\mathbf{v}_3 = \mathbf{p}_3 - \frac{\langle \mathbf{p}_3, \mathbf{v}_1 \rangle}{\|\mathbf{v}_1\|^2} \mathbf{v}_1 - \frac{\langle \mathbf{p}_3, \mathbf{v}_2 \rangle}{\|\mathbf{v}_2\|^2} \mathbf{v}_2 = x^2 - \frac{\frac{1}{3}}{1} (1) - \frac{\frac{1}{12}}{\frac{1}{12}} \left(-\frac{1}{2} + x\right) = x^2 - \frac{1}{3} + \frac{1}{2} - x = \frac{1}{6} - x + x^2$$

$$\|\mathbf{v}_3\|^2 = \langle \mathbf{v}_3, \mathbf{v}_3 \rangle = \int_0^1 \left(\frac{1}{6} - x + x^2 \right)^2 dx = \left(\frac{1}{36}x - \frac{1}{6}x^2 + \frac{4}{9}x^3 - \frac{1}{2}x^4 + \frac{1}{5}x^5 \right) \Big|_0^1 = \frac{1}{180}$$

$$\|\mathbf{v}_3\| = \sqrt{\frac{1}{180}} = \frac{1}{6\sqrt{5}}$$

The orthonormal basis is

$$\mathbf{q}_1 = \frac{\mathbf{v}_1}{\|\mathbf{v}_1\|} = \frac{1}{1} = 1;$$

$$\mathbf{q}_2 = \frac{\mathbf{v}_2}{\|\mathbf{v}_2\|} = \frac{-\frac{1}{2} + x}{\frac{1}{2\sqrt{3}}} = 2\sqrt{3} \left(-\frac{1}{2} + x \right) = \sqrt{3}(-1 + 2x);$$

$$\mathbf{q}_3 = \frac{\mathbf{v}_3}{\|\mathbf{v}_3\|} = \frac{\frac{1}{6} - x + x^2}{\frac{1}{6\sqrt{5}}} = 6\sqrt{5} \left(\frac{1}{6} - x + x^2 \right) = \sqrt{5}(1 - 6x + 6x^2).$$

44. The reduced row echelon form of the matrix is $\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{bmatrix}$ therefore the three column vectors of the

original matrix, $\mathbf{u}_1 = (6, 2, -2, 6)$, $\mathbf{u}_2 = (1, 1, -2, 8)$, and $\mathbf{u}_3 = (-5, 1, 5, -7)$ form a basis for the column space. Applying the Gram-Schmidt process yields an orthogonal basis $\{\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3\}$:

$$\mathbf{v}_1 = \mathbf{u}_1 = (6, 2, -2, 6)$$

$$\begin{aligned} \mathbf{v}_2 &= \mathbf{u}_2 - \frac{\langle \mathbf{u}_2, \mathbf{v}_1 \rangle}{\|\mathbf{v}_1\|^2} \mathbf{v}_1 = (1, 1, -2, 8) - \frac{6+2+4+48}{36+4+4+36} (6, 2, -2, 6) = (1, 1, -2, 8) - \frac{3}{4} (6, 2, -2, 6) \\ &= \left(-\frac{7}{2}, -\frac{1}{2}, -\frac{1}{2}, \frac{7}{2} \right) \end{aligned}$$

$$\begin{aligned} \mathbf{v}_3 &= \mathbf{u}_3 - \frac{\langle \mathbf{u}_3, \mathbf{v}_1 \rangle}{\|\mathbf{v}_1\|^2} \mathbf{v}_1 - \frac{\langle \mathbf{u}_3, \mathbf{v}_2 \rangle}{\|\mathbf{v}_2\|^2} \mathbf{v}_2 = (-5, 1, 5, -7) - \frac{-30+2-10-42}{36+4+4+36} (6, 2, -2, 6) - \frac{\frac{35}{2} - \frac{1}{2} - \frac{5}{2} - \frac{49}{2}}{\frac{49}{4} + \frac{1}{4} + \frac{1}{4} + \frac{49}{4}} \left(-\frac{7}{2}, -\frac{1}{2}, -\frac{1}{2}, \frac{7}{2} \right) \\ &= (-5, 1, 5, -7) + (6, 2, -2, 6) + \frac{2}{5} \left(-\frac{7}{2}, -\frac{1}{2}, -\frac{1}{2}, \frac{7}{2} \right) = \left(-\frac{2}{5}, \frac{14}{5}, \frac{14}{5}, \frac{2}{5} \right) \end{aligned}$$

45. Let $\mathbf{u}_1 = (1, 2)$, $\mathbf{u}_2 = (-1, 3)$, $\mathbf{q}_1 = \left(\frac{1}{\sqrt{5}}, \frac{2}{\sqrt{5}} \right)$, and $\mathbf{q}_2 = \left(-\frac{2}{\sqrt{5}}, \frac{1}{\sqrt{5}} \right)$. A QR-decomposition of the matrix A is formed by the given matrix Q and the matrix

$$R = \begin{bmatrix} \langle \mathbf{u}_1, \mathbf{q}_1 \rangle & \langle \mathbf{u}_2, \mathbf{q}_1 \rangle \\ 0 & \langle \mathbf{u}_2, \mathbf{q}_2 \rangle \end{bmatrix} = \begin{bmatrix} \frac{1}{\sqrt{5}} + \frac{4}{\sqrt{5}} & -\frac{1}{\sqrt{5}} + \frac{6}{\sqrt{5}} \\ 0 & \frac{2}{\sqrt{5}} + \frac{3}{\sqrt{5}} \end{bmatrix} = \begin{bmatrix} \sqrt{5} & \sqrt{5} \\ 0 & \sqrt{5} \end{bmatrix}.$$

46. Let $\mathbf{u}_1 = (1, 0, 1)$, $\mathbf{u}_2 = (2, 1, 4)$, $\mathbf{q}_1 = \left(\frac{1}{\sqrt{2}}, 0, \frac{1}{\sqrt{2}} \right)$, and $\mathbf{q}_2 = \left(-\frac{1}{\sqrt{3}}, \frac{1}{\sqrt{3}}, \frac{1}{\sqrt{3}} \right)$. A QR-decomposition of the matrix A is formed by the given matrix Q and the matrix

$$R = \begin{bmatrix} \langle \mathbf{u}_1, \mathbf{q}_1 \rangle & \langle \mathbf{u}_2, \mathbf{q}_1 \rangle \\ 0 & \langle \mathbf{u}_2, \mathbf{q}_2 \rangle \end{bmatrix} = \begin{bmatrix} \frac{1}{\sqrt{2}} + 0 + \frac{1}{\sqrt{2}} & \frac{2}{\sqrt{2}} + 0 + \frac{4}{\sqrt{2}} \\ 0 & -\frac{2}{\sqrt{3}} + \frac{1}{\sqrt{3}} + \frac{4}{\sqrt{3}} \end{bmatrix} = \begin{bmatrix} \sqrt{2} & 3\sqrt{2} \\ 0 & \sqrt{3} \end{bmatrix}.$$

47. Let $\mathbf{u}_1 = (1, 0, 1)$, $\mathbf{u}_2 = (0, 1, 2)$, $\mathbf{u}_3 = (2, 1, 0)$, $\mathbf{q}_1 = \left(\frac{1}{\sqrt{2}}, 0, \frac{1}{\sqrt{2}} \right)$, $\mathbf{q}_2 = \left(-\frac{1}{\sqrt{3}}, \frac{1}{\sqrt{3}}, \frac{1}{\sqrt{3}} \right)$, and

$\mathbf{q}_3 = \left(\frac{1}{\sqrt{6}}, \frac{2}{\sqrt{6}}, -\frac{1}{\sqrt{6}}\right)$. A QR-decomposition of the matrix A is formed by the given matrix Q and the matrix

$$R = \begin{bmatrix} \langle \mathbf{u}_1, \mathbf{q}_1 \rangle & \langle \mathbf{u}_2, \mathbf{q}_1 \rangle & \langle \mathbf{u}_3, \mathbf{q}_1 \rangle \\ 0 & \langle \mathbf{u}_2, \mathbf{q}_2 \rangle & \langle \mathbf{u}_3, \mathbf{q}_2 \rangle \\ 0 & 0 & \langle \mathbf{u}_3, \mathbf{q}_3 \rangle \end{bmatrix} = \begin{bmatrix} \frac{1}{\sqrt{2}} + 0 + \frac{1}{\sqrt{2}} & 0 + 0 + \frac{2}{\sqrt{2}} & \frac{2}{\sqrt{2}} + 0 + 0 \\ 0 & 0 + \frac{1}{\sqrt{3}} + \frac{2}{\sqrt{3}} & -\frac{2}{\sqrt{3}} + \frac{1}{\sqrt{3}} + 0 \\ 0 & 0 & \frac{2}{\sqrt{6}} + \frac{2}{\sqrt{6}} + 0 \end{bmatrix} = \begin{bmatrix} \sqrt{2} & \sqrt{2} & \sqrt{2} \\ 0 & \sqrt{3} & -\frac{1}{\sqrt{3}} \\ 0 & 0 & \frac{4}{\sqrt{6}} \end{bmatrix}.$$

48. Let $\mathbf{u}_1 = (1, 1, 0)$, $\mathbf{u}_2 = (2, 1, 3)$, $\mathbf{u}_3 = (1, 1, 1)$, $\mathbf{q}_1 = \left(\frac{1}{\sqrt{2}}, \frac{1}{\sqrt{2}}, 0\right)$, $\mathbf{q}_2 = \left(\frac{\sqrt{2}}{2\sqrt{19}}, -\frac{\sqrt{2}}{2\sqrt{19}}, \frac{3\sqrt{2}}{\sqrt{19}}\right)$, and

$\mathbf{q}_3 = \left(-\frac{3}{\sqrt{19}}, \frac{3}{\sqrt{19}}, \frac{1}{\sqrt{19}}\right)$. A QR-decomposition of the matrix A is formed by the given matrix Q and the matrix

$$R = \begin{bmatrix} \langle \mathbf{u}_1, \mathbf{q}_1 \rangle & \langle \mathbf{u}_2, \mathbf{q}_1 \rangle & \langle \mathbf{u}_3, \mathbf{q}_1 \rangle \\ 0 & \langle \mathbf{u}_2, \mathbf{q}_2 \rangle & \langle \mathbf{u}_3, \mathbf{q}_2 \rangle \\ 0 & 0 & \langle \mathbf{u}_3, \mathbf{q}_3 \rangle \end{bmatrix} = \begin{bmatrix} \frac{1}{\sqrt{2}} + \frac{1}{\sqrt{2}} + 0 & \frac{2}{\sqrt{2}} + \frac{1}{\sqrt{2}} + 0 & \frac{1}{\sqrt{2}} + \frac{1}{\sqrt{2}} + 0 \\ 0 & \frac{2\sqrt{2}}{2\sqrt{19}} - \frac{\sqrt{2}}{2\sqrt{19}} + \frac{9\sqrt{2}}{\sqrt{19}} & \frac{\sqrt{2}}{2\sqrt{19}} - \frac{\sqrt{2}}{2\sqrt{19}} + \frac{3\sqrt{2}}{\sqrt{19}} \\ 0 & 0 & -\frac{3}{\sqrt{19}} + \frac{3}{\sqrt{19}} + \frac{1}{\sqrt{19}} \end{bmatrix}$$

$$= \begin{bmatrix} \sqrt{2} & \frac{3}{\sqrt{2}} & \sqrt{2} \\ 0 & \frac{\sqrt{19}}{\sqrt{2}} & \frac{3\sqrt{2}}{\sqrt{19}} \\ 0 & 0 & \frac{1}{\sqrt{19}} \end{bmatrix}.$$

49. In partitioned form, $A = \begin{bmatrix} 1 & 0 & 1 \\ -1 & 1 & 1 \\ 1 & 0 & 1 \\ -1 & 1 & 1 \end{bmatrix} = [\mathbf{u}_1 \quad \mathbf{u}_2 \quad \mathbf{u}_3]$. By inspection, $\mathbf{u}_3 = \mathbf{u}_1 + 2\mathbf{u}_2$, so the column vectors

of A are not linearly independent and A does not have a QR-decomposition.

51. The proof of part (a) mirrors the proof of part (b) in the book. By Theorem 6.3.1, an orthogonal set of nonzero vectors in W is linearly independent. It follows from part (b) of Theorem 4.6.5 that this set can be enlarged to form a basis for W . Applying the Gram-Schmidt process (without the normalization step) will yield an enlarged orthogonal set (the original orthogonal set will not be affected).
52. If $\mathbf{v}_3 = \mathbf{0}$ then

$$\begin{aligned} \mathbf{u}_3 &= \frac{\langle \mathbf{u}_3, \mathbf{v}_1 \rangle}{\|\mathbf{v}_1\|^2} \mathbf{v}_1 + \frac{\langle \mathbf{u}_3, \mathbf{v}_2 \rangle}{\|\mathbf{v}_2\|^2} \mathbf{v}_2 \\ &= \frac{\langle \mathbf{u}_3, \mathbf{v}_1 \rangle}{\|\mathbf{v}_1\|^2} \mathbf{u}_1 + \frac{\langle \mathbf{u}_3, \mathbf{v}_2 \rangle}{\|\mathbf{v}_2\|^2} \mathbf{u}_2 - \frac{\langle \mathbf{u}_3, \mathbf{v}_2 \rangle \langle \mathbf{u}_2, \mathbf{v}_1 \rangle}{\|\mathbf{v}_2\|^2 \|\mathbf{v}_1\|^2} \mathbf{u}_1 \\ &= \left(\frac{\langle \mathbf{u}_3, \mathbf{v}_1 \rangle}{\|\mathbf{v}_1\|^2} - \frac{\langle \mathbf{u}_3, \mathbf{v}_2 \rangle \langle \mathbf{u}_2, \mathbf{v}_1 \rangle}{\|\mathbf{v}_2\|^2 \|\mathbf{v}_1\|^2} \right) \mathbf{u}_1 + \frac{\langle \mathbf{u}_3, \mathbf{v}_2 \rangle}{\|\mathbf{v}_2\|^2} \mathbf{u}_2 \end{aligned}$$

making $\mathbf{u}_3$ a linear combination of $\mathbf{u}_1$ and $\mathbf{u}_2$, which contradicts the assumption of the linear independence of $\{\mathbf{u}_1, \mathbf{u}_2, \mathbf{u}_3\}$.

53. The diagonal entries of R are $\langle \mathbf{u}_i, \mathbf{q}_i \rangle$ for $i = 1, 2, \dots, n$, where $\mathbf{q}_i = \frac{\mathbf{v}_i}{\|\mathbf{v}_i\|}$ is the normalization of a vector $\mathbf{v}_i$ that is the result of applying the Gram-Schmidt process to $\{\mathbf{u}_1, \mathbf{u}_2, \dots, \mathbf{u}_n\}$. Thus, $\mathbf{v}_i$ is $\mathbf{u}_i$ minus a linear combination of the vectors $\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_{i-1}$, so $\mathbf{u}_i = \mathbf{v}_i + k_1\mathbf{v}_1 + k_2\mathbf{v}_2 + \dots + k_{i-1}\mathbf{v}_{i-1}$. Thus, $\langle \mathbf{u}_i, \mathbf{v}_i \rangle = \langle \mathbf{v}_i, \mathbf{v}_i \rangle$ and $\langle \mathbf{u}_i, \mathbf{q}_i \rangle = \left\langle \mathbf{u}_i, \frac{\mathbf{v}_i}{\|\mathbf{v}_i\|} \right\rangle = \frac{1}{\|\mathbf{v}_i\|} \langle \mathbf{v}_i, \mathbf{v}_i \rangle = \|\mathbf{v}_i\|$. Since each vector $\mathbf{v}_i$ is nonzero, each diagonal entry of R is nonzero.

55. (b) The range of T is W ; the kernel of T is $W^\perp$.

True-False Exercises

- (a) False. For example, the vectors $(1, 0)$ and $(1, 1)$ in $\mathbb{R}^2$ are linearly independent but not orthogonal.
- (b) False. The vectors must be nonzero for this to be true.
- (c) True. A nontrivial subspace of $\mathbb{R}^3$ will have a basis, which can be transformed into an orthonormal basis with respect to the Euclidean inner product.
- (d) True. A nonzero finite-dimensional inner product space will have finite basis which can be transformed into an orthonormal basis with respect to the inner product via the Gram-Schmidt process with normalization.
- (e) False. $\text{proj}_W \mathbf{x}$ is a vector in W .
- (f) True. Every invertible $n \times n$ matrix has a QR -decomposition.

6.4 Best Approximation; Least Squares

$$1. \quad A = \begin{bmatrix} 1 & -1 \\ 2 & 3 \\ 4 & 5 \end{bmatrix}; \quad A^T A = \begin{bmatrix} 1 & 2 & 4 \\ -1 & 3 & 5 \end{bmatrix} \begin{bmatrix} 1 & -1 \\ 2 & 3 \\ 4 & 5 \end{bmatrix} = \begin{bmatrix} 21 & 25 \\ 25 & 35 \end{bmatrix}; \quad A^T \mathbf{b} = \begin{bmatrix} 1 & 2 & 4 \\ -1 & 3 & 5 \end{bmatrix} \begin{bmatrix} 2 \\ -1 \\ 5 \end{bmatrix} = \begin{bmatrix} 20 \\ 20 \end{bmatrix};$$

$$\text{The associated normal equation is } \begin{bmatrix} 21 & 25 \\ 25 & 35 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 20 \\ 20 \end{bmatrix}.$$